Renormalized Connection for Scale-Preferred Object Detection in Satellite Imagery

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***Abstract*— Satellite imagery, due to its long-range imaging, brings with it a variety of scale-preferred tasks, such as the detection of tiny/small objects, making the precise localization and detection of small objects of interest a challenging task. In this article, we design a knowledge discovery network (KDN) to implement the renormalization group theory in terms of efficient feature extraction (FE). Renormalized connection (RC) on the KDN enables "synergistic focusing" of multiscale features. Based on our observations of KDN, we abstract a class of RCs with different connection strengths, called *n*21C, and generalize it to feature pyramid network (FPN)-based multibranch detectors. In a series of FPN experiments on the scale-preferred tasks, we found that the "divide-and-conquer" idea of FPN severely hampers the detector's learning in the right direction due to the large number of large-scale negative samples and interference from background noise. Moreover, these negative samples cannot be eliminated by the focal loss function. The RCs extends the multilevel feature's "divide-and-conquer" mechanism of the FPN-based detectors to a wide range of scale-preferred tasks, and enables synergistic effects of multilevel features on the specific learning goal. In addition, interference activations in two aspects are greatly reduced and the detector learns in a more correct direction. Extensive experiments of 17 well-designed detection architectures embedded with *n*21Cs on five different levels of scale-preferred tasks validate the effectiveness and efficiency of the RCs. Especially the simplest linear form of RC—E421C performs well in all tasks, and it satisfies the scaling property of renormalization group theory. All experiments can be trained and tested on a graphics card with 8 GB of video memory, which greatly enhances the applicability of our methodology. We hope that our approach will transfer a large number of well-designed detectors from the computer vision community to the remote sensing community. Datasets and codes will be available at: https://github.com/rabbitme/**

***Index Terms*— Feature extraction (FE), knowledge discovery network (KDN), object detection, remote sensing, renormalized connection (RC), satellite image object detection, small object detection.**

I. Introduction

With the rapid advancement of deep learning technology and hardware, the application scope of intelligent interpretation of satellite imagery has gradually expanded. In the field of remote sensing, the accuracy of automatic detection of large objects has become comparable to that of human experts [1], [2] and has been applied to aspects of people's production and life. However, there are a large number of small objects in satellite imagery [3], [4], and there is still room to improve the accuracy of small object detection. In generic object detection community, there are a huge number of well-designed algorithms including higher capacity backbone networks [5], [6], [7] and feature extractors with rich feature space [8], [9], [10], [11], [12], aligned head network [13] to cope with performance enhancement problems. Compared with the scale-diversified object detection tasks, small object detection tasks can be seen as a scale preference problem.[1]

Recently, the remote sensing community has proposed well-designed tasks and solutions to concentrate on small object detection [14]. For accurate object recognition, mining useful features in a huge space is an important direction. Jia et al. [15] investigated in detail a series of feature mining methods for hyperspectral image classification, including feature selection and feature extraction (FE) methods. Chen et al. [16] presented a regularized deep FE method and verified its validity. Ma et al. [17] designed a spectral–spatial feature mining framework for noise-robust FE. Context features are another crucial piece of information. Li et al. [18] raised a separate detector-based feature fusion (SDFF) architecture that performs interlayer feature fusion to incorporate multiscale context in the dynamic image pyramid of the large-size image. Luo et al. [19] first used a single point-based oriented bounding boxes (OBB) generation method for oriented object detection. Inspired by these effective works, we design feature mining methods to tackle difficult scale preference tasks.

For scale related problems, the most popular FE method is feature pyramid network (FPN) [8], which can simultaneously detect different scale objects in an end-to-end manner. The core design idea of FPN is "divide-and-conquer"; that is, each feature level in the pyramid performs its own function, to detect objects within a certain scale range. Based on this, path aggregation FPN (PAFPN) [9] extended the number of connection layers of the feature extractor by downsampling the largest feature map output from the FPN. It is straightforward to use FPN-based detectors to tackle scale-preferred problems. However, when we study scale-preferred tasks, such as the tiny/small object detection tasks (Fig. 1), the direct use of

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[1]Scale-preferred tasks are tasks with a scale preference, such as small object detection or any object detection tasks under remote sensing imagery acquired at a fixed viewing angle, where objects of the same class are of similar size.

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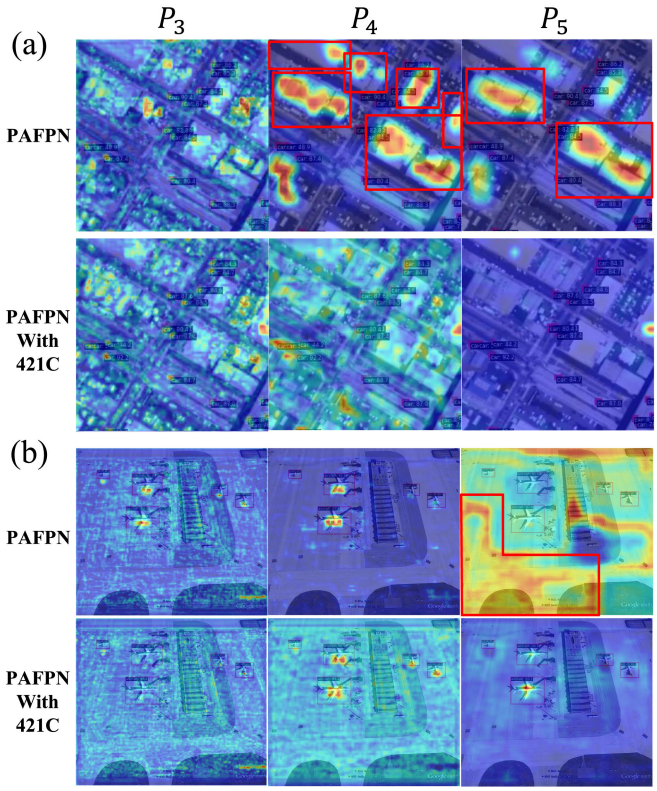


Fig. 1. Saliency maps of YOLOv8-PAFPN describe the importance of features. The FPN series of extractors with focal loss are unable to adapt to difficult scale-preferred tasks. The initial "divide-and-conquer" mechanism for different pyramid levels brings a large number of interfering activations (areas with red activation values) for tiny object detection tasks. This figure shows the saliency maps of the three pyramid level feature activations of the feature extractor PAFPN for two types of scale preference tasks. (a) Purely tiny object detection task, we can see that the P_4- and P_5-level feature maps highlight the blockbuster background objects that have similar visual appearances (color and shape) but with different scales and object classes (in red boxes). This will greatly overwhelm the detection focus of P_3. (b) Small object detection task, where the P_3 and P_4 levels all focus on the same-sized objects which ought to be detected by P_4. The P_5 level, however, is more concerned with the large irrelevant regions in the background. All the observations suggest that the signal-to-noise ratio of the FPNs used for the scale-preferred tasks is lower than the scale-diversified tasks because the huge amount of interfering activations overwhelmingly prevent the previous levels [(a) P_3 and (b) P_3 and P_4] from focusing on small objects as the objective.

the FPN detector leads to serious problems with interfering activations, both in terms of large numbers of negative samples and background noises, caused by the "divide-and-conquer" idea and the multisubtask parallel learning of different feature levels of FPN.

Notably, Fig. 1 shows the saliency map of the one-stage detector YOLOv8-PAFPN coupled with focal loss function [20] after 500 epochs of training. As we know, the focal loss [21] aims to address the foreground–background class imbalance problem of one-stage detectors on scale-diversified datasets. However, on scale-preferred datasets, a large number of negative samples are actively generated which produce large activation values that cannot be eliminated during training. These two aspects of the observations suggest that there is still a noticeable problem in scale-preferred tasks. That is, the high-level features of FPN propose a large number of negative samples, and the focal loss cannot eliminate these interfering activations during training.

In small object detection tasks, if the interfering activations in the large object detection branch are eliminated, there will be no interfering gradient information when running backpropagation. The learning process will then be redirected in the right direction. Therefore, we aim to address the problem that branches assigned fewer positive samples produce interference activations (the distribution of activation values can be visualized from the saliency maps) in tasks with only small objects and, more generally, in tasks with scale preference. If this problem is solved, then a large number of methods designed for tasks with diverse scales can be used to solve the scale-preferred problem, thus greatly expanding the set of solutions to various scale-preferred problems.

In this article, because the focal loss cannot eliminate the actively generated large number of negative samples by FPN's P_5 level on scale-preferred tasks, we consider using renormalization group theory to guide the detector to focus on hard samples. At first, we design a knowledge discovery network (KDN) on a single-branch detector to find the relationship between different stage features in the feature extractor and the function of each stage for the ultimate detection task. Without the need to design a novel positive–negative sample assignment method for the scale-preferred task, we start from the network structure design (KDN) and bridge a renormalized connection (RC) between the feature extractor and the detection head, directing the multilevel outputs of the feature extractor to be correctly used for the feature supply of the detection head.

After a series of experiments of KDN on different small object detection datasets, we find that although the stages in ResNet, except the C_3 stage, are not directly involved in the detection task, each stage produces different levels of features that are both independent of and complementary to each other. This is consistent with the idea that different levels of features have their own roles in the FPN. Inspired by this observation, we generalize the RCs obtained by KDN to the more general case of FPN-based multibranch detectors. To mathematically abstract this RC method, we construct a set of renormalized coefficients, called connection strengths, for a set of mutually independent bases on the feature space generated by the feature extractor. We name this type of RC according to the coefficient ratios, as n21C. Among them, the simplest uniformly RC coefficients conform to the scaling property in renormalization group theory, with coefficient ratios of 421, denoted by 421C.

The n21Cs represent a class of RCs, in which the coefficient ratios can be varied according to the task requirements, and the optimal coefficient ratios for each specific task can be found by a grid search or a random search. When used for tasks with specific scale preferences, we find analytically and experimentally in this article that very small, small, and scale-diversified tasks can use the 421 coefficient ratios to renormalize the connections of three or more different detection branches. The 421Cs eliminate interference activations on the large object detection branch and achieve the goal of detection branches not interfering with each other and learning collaboratively on the task of focusing on detecting small objects. The RCs extend the idea of the multilevel feature's "divide-and-conquer" mechanism of the FPN-based

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detectors to the scale-preferred tasks and enable synergistic effects of multilevel features on the specific learning goals. It greatly eliminates interfering activations from the branches assigned fewer positive samples of the FPN that generate a large number of negative samples and background noise before they are fed into the detection network. The n21Cs assign appropriate subtasks to multiple detection branches for scale-preferred tasks. In a word, RCs in multibranch detectors enable an efficient mechanism for processing multiple subtasks in parallel and focusing on the dominant subtask. (The task assigned to the detection branch, where the RC is inserted is the dominant task.)

Facing the practical application requirements of remote sensing satellite videos, we have specially designed a purely tiny object detection problem with the following challenges.

1) *Small Size:* All the objects to be detected in the dataset are extremely small in size. Small objects account for a small proportion of the scene, a small area, and a small number of pixel points.
2) *Low Image Ground Resolution:* When the image ground resolution is low, the boundaries of small objects are not sharp, and most of the features disappear, making it difficult to distinguish them and resulting in missed detections.
3) *High Density of Distribution:* If the objects to be detected have a high distribution density, it is easy to lead to missed detection.
4) *Low Contrast:* Due to the long distance of satellite image shooting, small objects are not rich in color and texture and have low contrast with the background, which can easily lead to detection errors.
5) *Interfering Objects:* Objects with similar top surfaces are scattered throughout the scene and can easily cause confusion.

A new tiny object detection dataset, called the IPIU dataset, is being created to address all these challenges. This helps us to construct the task with the highest scale-preferred in this article. In addition, we conducted experiments on four different datasets with different ranges of scale distributions.

We validate the performance of the RCs on 17 well-designed detection architectures with three dominant backbone networks as well as feature extractors with different connection layers and densities, on four satellite imagery datasets and a natural small object detection dataset.[2] Extensive experiments demonstrate the effectiveness and high efficiency of the RCs. On a variety of scale-preferred tasks, n21Cs eliminate two aspects of interfering activations and allow the network to fully focus on the target to be detected, significantly improving learning efficiency and detection performance. Since all experiments can be trained and tested on a graphics card with limited video memory, the range of applications is greatly enhanced!

In summary, our contributions are as follows.

1) We propose a KDN on a single-branch detector to implement the renormalization group theory on FE network and find that the feature extractor produces independent multilevel features and these different layers of features act synergistically on the detection task which is consistent with the "each in its own way" idea of multilevel FPN. We refer to the mechanism of KDN's RC as "synergistic focusing."[3]
2) Based on the observations derived from KDN, we mathematically abstract a class of RCs with different coefficient ratios called n21Cs and generalize them to the FPN-based multibranch detectors, thereby significantly eliminating negative samples and interfering activations in various scale-preferred tasks. For FPN-based n21Cs, they implement both "divide-and-conquer" and "synergistic focusing" mechanisms as well as appropriate subtask assignment for parallel training of multiple detection branches on scale-preferred tasks. The RCs are customized connection methods.
3) We create a difficult scale-preferred task on satellite imagery applications, that is IPIU dataset, a purely tiny object detection dataset, and provide a benchmark to fairly compare the performance of different detection algorithms.
4) Extensive experimental results, including mean average precision (mAP) and average recall (AR), average precision (AP) and loss curves, saliency maps, and visualizations, have verified that the RC approach can renormalize the information flow in the forward and back propagation phases of network learning and adjust the learning process to a more correct direction.

The remainder of this article is organized as follows. Section II summarizes typical connection modules and state-of-the-art satellite image object detectors. Section III introduces the application of renormalization group theory to FE, i.e., the KDN. Based on the observations of the network, we construct a new type of connection called an RC. It can be generalized to a variety of detectors. Section IV reports the experimental setup, performance comparisons, visualization results, and discussions on five representative scale-preferred or scale-diversified datasets. Finally, Section V concludes this article and identifies future work.

II. Related Works

In this section, we review representative connection modules in the history of deep networks and survey novel satellite image detection methods. The connection strategies [5], [22], [23] in network architecture design have made conspicuous performance gains in a variety of computer vision tasks. Research on satellite applications from different perspectives has flourished in the community.

A. Connection Modules in Deep Networks

It is mentioned in ResNet [5] that deep networks cannot automatically learn the structure of identity mapping and need to manually add residual connections. Consequently, in addition to designing the basic module of the backbone network, in the detection task, researchers have also worked on designing feature extractors with different connection methods. As shown in Fig. 2, we draw the RC (as known as n421C of the

[2]Includes one newly released high-difficulty scale-preferred dataset, one scale-preferred dataset with multiple classes and complex scenarios, and three datasets with more varied scales and different object categories of interest.

[3]Multiscale features collaborate and do not interfere with each other, focusing on the same learning objective (in scale-preferred problems, where objects have a preferred scale distribution, e.g. small objects in SOD).

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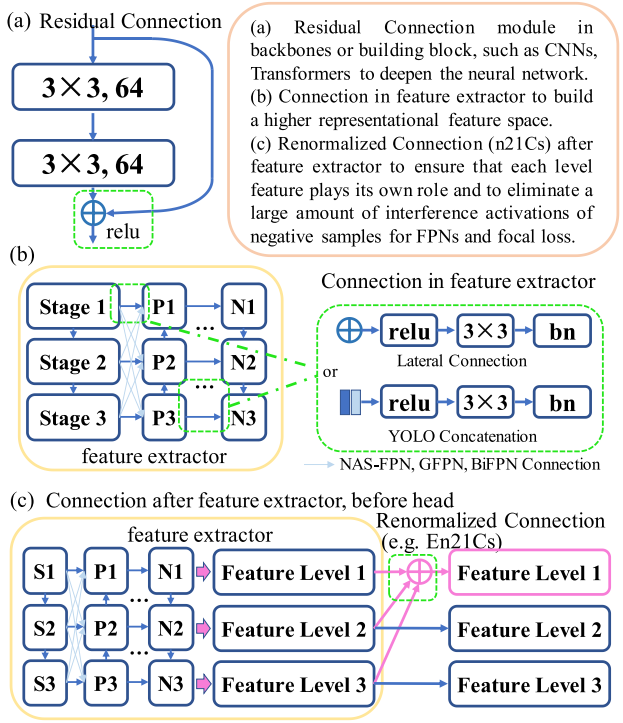


Fig. 2. Three types of feature connection methods. (a) Residual connection is designed to deepen the network by introducing a nonlinear computation after summating the input and building block features, i.e., the rectified linear unit (ReLU). (b) Gathers the design structures of different feature extractors with different connection layers and connection densities, such as GiraffeDet [24], RTMDet [25], neural architecture search (NAS)-FPN [10], BiFPN [11], and GFPN [26]. Lateral connection [8] is an independent connection module including summation or concatenation, ReLU layer, 3 × 3 convolutional layer, and batch normalization layer. In addition to lateral connections for feature maps of the same size, there are also connections for feature maps of different sizes (indicated by light blue arrows). (c) RC (n21Cs), which is used to renormalize the output of the feature extractor and form a new input to the head net. This is an example of economical n21C, embedded in Feature Level 1 only. It only uses the output features of the designed feature extractor as inputs and does not change the internal connection structure of the feature extractors. Moreover, this kind of RC is a linear connection structure and does not contain additional nonlinear operations or any learnable layer (except for the projection operation) and normalization layers yet works well.

generalized RC) with the residual connection, the connection of multiple FPN-based feature extractors.

Differences between RC and various existing connection methods.

1) *Different Scope of Action:* Residual connection is used to deepen the backbone network, feature extractor, or any building block in a variety of deep networks. Lateral connection is used for feature fusion within the feature extractor. Also, YOLO concatenation from v3 [27] to v8 [20] is used for feature fusion in each base module. The RC is used for feature renormalization for the feature extractor to provide a "synergistic focusing" feature pyramid for the detection head and to eliminate interfering activations for a large variety of scale-preferred tasks. RC is a bridge between the feature extractor and the detection head, directing the multilevel outputs of the feature extractor to be correctly used for the feature supply of the embedded branch in the detection network. The connection strength, embedding position (one or more detection branches), and the number of RCs can be flexibly set to meet different task requirements.
2) *Different Purposes:* The residual connection and lateral connection methods are used to increase the capacity of the model. In contrast, the RC is used to renormalize the information flow at all stages of network learning (rearranging the feature flow in the forward propagation phase and reorienting the gradient flow in the backpropagation phase), adjusting the learning process in a more correct direction.
3) The complexity of the connections is different. The residual connection and lateral connection work well with nonlinear operations involving nonlinear activation units, batch normalization layers, or new convolutional layers. These nonlinear connection methods can be considered as a completely new module. The RC produces a very good renormalizing effect using only linear connections. However, adding nonlinearities to the linear RC, such as the 1 × 1 conv and the normalization layer, can significantly reduce performance.

The difference between the RC and the bottom–up path in PAFPN [9] is that the inputs to the economical RC (E421C, only for the small object detection branch corresponding to the P_3 level in FPN) are the three direct outputs of the top–down path in the FPN, whereas the bottom–up path in PAFPN uses only one of the features of the top–down path in the FPN (P_3) as the input to the small object detection branch. Moreover, PAFPN acts as a feature extractor and the RC can be used on top of it as a bridge to connect the feature extractor and head.

The RC has a different scope of application and functionality compared to the region of interest (RoI) extractors [9], [28] (feature fusion methods for RoIs). Since only the two-stage object detector contains an RoI pooling/RoI align layer, the RoI feature fusion methods [9], [28] can only be used on two-stage detectors. In contrast, the RCs can be used with an RoI extractor on two-stage detectors and greatly eliminate the inference activations of one-stage detectors. The RoI extractors are all designed for scale-diversified datasets, fusing only features within the RoI and only on two-stage detectors. The RC is a connection between the feature extractor and head to renormalize information flow for various detectors.

On top of the broader and more powerful detection architectures built by various connection methods, RCs can further improve the accuracy of scale-preferred detection tasks. Most importantly, the simplest RC, i.e., E421C, can make the above connection methods work well in a variety of scale-preferred tasks, thus greatly extending their range of applications.

RCs achieve the design idea that all the detection branches play their own roles without interfering with each other, focusing on the dominant task. RC inherits the "divide-and-conquer" and multisubtask parallel learning mechanisms of FPN-based detectors, and brings a "synergistic focusing" mechanism to detectors to solve scale-preferred tasks.

B. Satellite Imagery Object Detection

For small object detection, Cao et al. [29] used physical simulation images as prior knowledge to extract the target's key geometric features through a dynamic matching method. Yin et al. [30] proposed a lightweight hybrid attention block for ship detection. Wang et al. [4] introduced a random projection feature for characterizing blocks into feature

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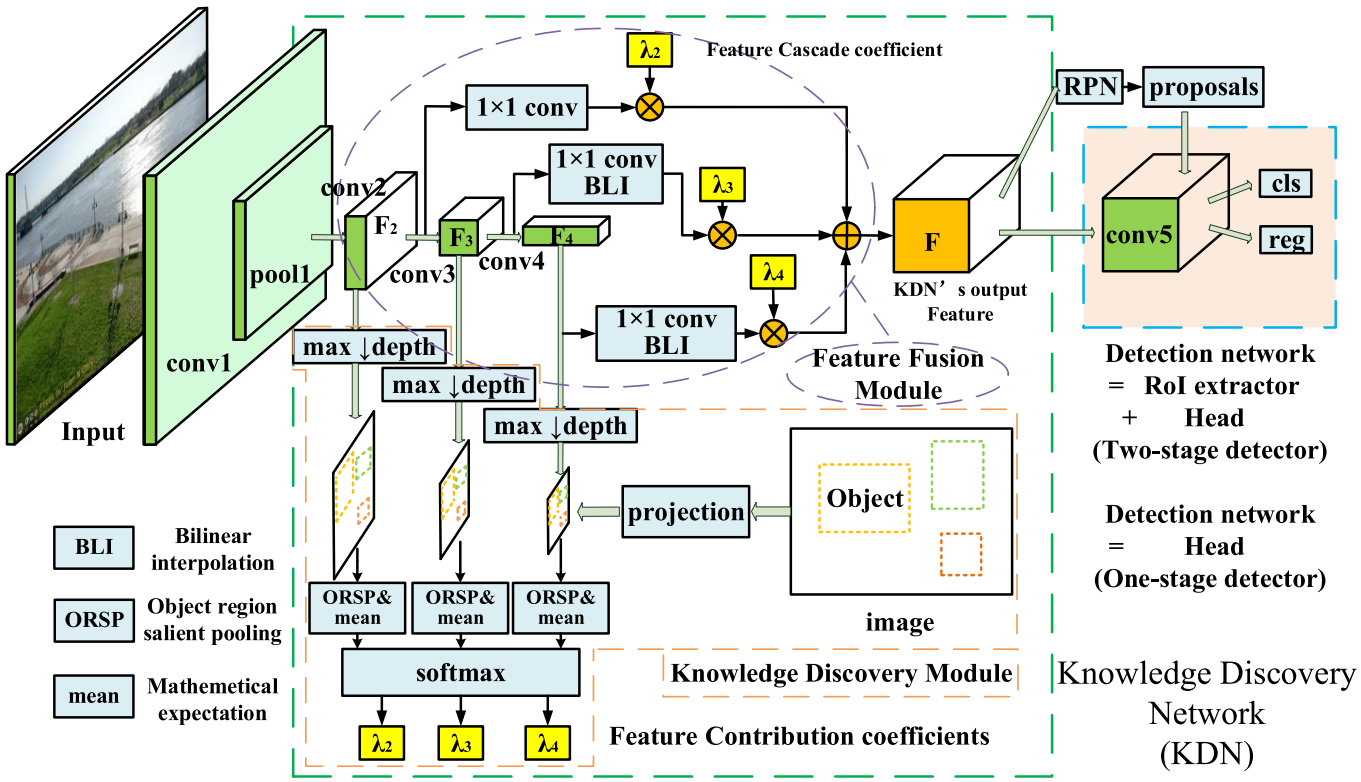


Fig. 3. Framework of the KDN.

vectors to solve the problems with dynamic background and scale variation targets. Yuan et al. [31] proposed a classifier for vehicle classification under foggy weather conditions. Yuan et al. [32] captured context to enhance classification scores by enlarging receptive field regions and designing head networks. Zhang et al. [33] used super-resolution and cross-modality information fusion.

Moreover, recently, satellite remote sensing images have plenty of applications, such as cloud detection [34], [35], multiclass object counting, and localization [36].

III. Methodology

In this section, we present methods to cope with the scale-preferred tasks. First, we introduce renormalization group theory for FE. Specifically, we construct a renormalized feature distribution for various feature extractors and redirect the learning for the detector. We implement the renormalization group theory on comprehensive FE by designing a KDN. We then use the observations of the KDN to build a new type of connection, called an RC, denoted by n21C ($n \in R$). This n21C method can be generalized to a variety of detectors. The simplest and most generalizable type of RC is the economical 421 connection. Except for the 421C, we have constructed the 521C and the complete form of RC to depict the RCs from a broad perspective. Now, let us take a look at the promising renormalization methods.

A. Renormalization Group Method for FE

A common feature of problems where renormalization appears is the presence of infinitely many particles (real or virtual) in the interaction between them, often with an emergent behavior that can be very different at different energy scales [37]. For example, we use a tree to illustrate. The renormalization method focuses on the self-similarity of the tree since each branch of the tree is similar to the traits of the whole tree, the branches are similar to smaller branches, and so on. It is the features hidden in these iterative operations that the renormalization group [38] wants to extract, hoping to unearth the relationships between the system at different scales.

On a deep neural network, as the network moves from lower to higher layers, the features it extracts become more and more abstract and involved in the nature of the "whole." At a deeper level, deep neural networks are not simply a stack of neural networks, but can actually find some hidden symmetries and constraints in the data. This means that the features extracted by deep neural networks will be "slow variables" that are not affected by noise, which is consistent with the idea of renormalization, that is, mutually independent variables with appropriate degrees of freedom.

The basic physical idea underlying the renormalization group approach is that the many length or energy scales are locally coupled [39]. Inspired by the methodology of the renormalization group in statistical mechanics, we build a renormalization method for comprehensive FE. The aforementioned idea has resulted that there is a cascade effect on the whole system. In multiscale feature extractors, the different but contiguous feature scales also have a cascade effect. For example, the behavior of the feature stage C_3 in Faster region-based convolutional neural network (R-CNN) is responsible for detecting tasks and is directly sent to the head network

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are assumed to be primarily affected by the nearby feature stages, e.g., C_2 and C_4. The single-branch detector uses only single-scale features (one stage) for the detection task, and the remaining feature stages constitute the feature extractor of the detector. To verify that there is a cascade effect in the feature extractor, the single-branch detector is the best choice. Once the base model is selected, we build the cascade for the renormalization method.

There are two principal properties ([39] used features) of the feature cascade. The first property is scaling. The behavior of FE for intermediate feature stages tends to be identical except for a change of scale. We will verify it in Section III-B6. We focus here on describing the second property, the existence of amplification and deamplification as the cascade develops. That is, we have to determine the quantitative relationship and interaction between the contiguous feature cascade. Corresponding to a single-branch detector, we have to determine the extent to which the previous and subsequent feature stages affect the detection task. The amplification effect implies that there are complementary and synergistic effects between any successive two feature scales. The deamplification effect, on the other hand, indicates a mutual exclusivity and correlation between these features.

Next, we perform the renormalization group analysis of the detection system to understand the relationship between contiguous feature stages in the feature extractor.

B. KDN—the Implementation of Renormalization Group Theory in FE

We design a KDN to implement renormalization group theory in FE. The purpose of using renormalization group theory is to renormalize the output feature flow of the feature extractor to further guide the detector to find a suitable learning direction quickly. As mentioned above, the single-branch detector is the best choice to verify the cascade effect in the feature extractor. Therefore, we construct the KDN on a single-branch detector, as shown in Fig. 3.

The first step in renormalization is to determine the feature cascade consisting of features at different scales in the feature extractor. Each scale of features in the cascade represents an independent variable. The independent degrees of freedom for FE in single-branch detectors is 3. The renormalization group finds the relevant variables, and once we have combined all the relevant variables, we will have "a view of the mountains." The combined relevant variables are used to describe the features that govern the more comprehensive behavior of the system and are independent of noise and local fluctuations.

In the second step, we should calculate the amplification and deamplification factors λ_i for each scale ($i = 2, 3, 4$). This consists of an explicit statistical averaging over the knowledge that we have extracted from each information source in the feature cascade. The third step is to construct a renormalization group for the detection head.

For example, we use Faster R-CNN as a single-branch detector. The backbone provides three-stage features to compose an FE network. Thus, the feature cascade consists of F_2, F_3, and F_4 with a scale multiplier of 2. The amplification factor or deamplification factor is generated by mining the knowledge in the information source. For our task, the information sources are the objects in the dataset. The KDN translates the knowledge into an amplification or deamplification factor for each stage in the feature cascade. To obtain knowledge from all information sources, we need to traverse all the images in the dataset. Naturally, traversal can be achieved by training the network. After one epoch of training, all images are traversed once.

1) Architecture of KDN: KDN is located in the middle part between the feature extractor and the detection network.[4] The KDN consists of two parts: a knowledge discovery module (KDM) and a feature fusion module (FFM). The KDM executes the first and second steps of the renormalization method. The FFM generates the renormalization group for the detection network. After training, we can get the final factor set for inference, which determines the amplification or deamplification factor for each independent variable in the feature cascade.

2) Knowledge Discovery Module: The KDM extracts information from each object, then generates useful knowledge through statistical analysis, and finally produces the amplification or deamplification factor for each scale in the feature cascade. Because we only focus on small objects, after a series of downsampling operations, the remaining feature maps have a smaller width and height; the effective feature region of small objects also remains smaller. For example, a 4×4 pixel object in the original image is left with one effective pixel in the output feature map after two downsampling layers with stride 2. Therefore, we select the most salient feature value in the projection region of each feature scale, which is sufficient to represent the information of the object on the feature map.

First, the object's position is projected onto the feature map at each level. Then, an effective feature region of the object is determined for each scale in the feature cascade. The salient information of an object can be extracted from its effective feature region. Given an input image and a feature scale, the KDM extracts the salient information of each small object (on the projected feature region) and stores it in a stack. After collecting information about all objects of interest in the image, the KDM calculates the mathematical expectation of the salient stack to represent the global knowledge of all objects in the image mapped to that feature level, and from this determines an amplification factor or deamplification factor for each scale. If the factor value is greater than 1, it is determined as an amplification factor; otherwise, it is determined as a deamplification factor. The amplification factor corresponds to a relevant variable. A variable whose effect is deamplified is called an "irrelevant variable." The number of relevant variables is equal to the number of amplification factors λ_i greater than 1.[5] Finally, the relevant variables are combined to characterize the features that govern the more comprehensive behavior of the system for detection tasks.

[4]The detection network in this work specifies a head network with one or more branches rather than an entire detector, where each branch comprises a classification subnetwork and a regression subnetwork.

[5]In this article, we use a looser $\lambda_i \geqslant 1$ condition.

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For each image (has n objects) and its corresponding feature cascade (with degrees of freedom as 3), KDM performs the same calculation as follows.

Step 1: Reduce_max operation (returns the maximum value of each channel in the feature maps)

$$rm(F_l), (l = 2, 3, 4). \tag{1}$$

Step 2: Object region projection

$$\text{orp}_i(rm(F_l)), (i = 1, \ldots, n; l = 2, 3, 4). \tag{2}$$

Step 3: Object (projected) region salient pooling

$$\text{ORSP}_i(\text{orp}_i(rm(F_l))), (i = 1, \ldots, n; l = 2, 3, 4). \tag{3}$$

Step 4: Mathematical expectation calculation

$$\lambda_l^e = \frac{\sum_i \text{ORSP}_i(\text{orp}_i(rm(F_l)))}{n}, (i = 1, \ldots, n; l = 2, 3, 4). \tag{4}$$

Step 5: Softmax operation

$$\lambda_l = \frac{\exp(\lambda_l^e)}{\sum_j \exp(\lambda_j^e)}, (j, l = 2, 3, 4). \tag{5}$$

Step 6: If $\lambda_i \geqslant 1$, $(i = 2, 3, 4)$: returns an amplification factor; else: returns a deamplification factor.

Step 7: The amplification factor corresponds to a relevant variable in the feature cascade $\{F_l, l = 2, 3, 4\}$. Retaining all the relevant variables in the feature cascade as $\{F_i, i = np.\text{where}(\lambda_i \geqslant 1)\}$.[6]

3) Feature Fusion Module: As usual, the feature extractor has different "level" features of different scales and dimensions. The content of different "level" features is also different; some of them focus on representing details, and others focus on representing more abstract features. To capture comprehensive features of an object, it is beneficial to integrate different levels of features. The feature fusion module (FFM) aims to generate a more complete feature description for the detection network by integrating different "levels" of features that make up the feature cascade.

The input for each branch in the detection head has a fixed scale and dimension. Consequently, the KDN has to output features of fixed size. Features of different "levels" must be mapped to the same size and dimension. For example, if the detection network requires the input of features from the middle level, FFM should map the high-level features to the middle-level and low-level features to the middle level to ensure that features from all levels have the same size and dimension. Unlike the form of feature fusion that connects the layers within the feature extractor—direct summation or concatenation operations—we wish to fuse the features by means of a corresponding amplification or deamplification factor. The amplification or deamplification factor is the extent to which each "level" of features in the feature cascade contributes to the overall feature. After obtaining features of the same size and dimension in FFM, we use the amplification or deamplification factors generated by the KDM to output a comprehensive feature for detection as follows:

$$F = p(F_2) \times \lambda_2 + \text{BLI}(p(F_3)) \times \lambda_3 + \text{BLI}(p(F_4)) \times \lambda_4 \tag{6}$$

where $p()$ is the 1×1 conv projection.

The FFM generates the comprehensive features in Step 8 after Step 7 of the KDM.

Step 8: During training, the combination of all independent variables multiplied by the corresponding factors yields the comprehensive features as described in (6).

4) Final Factor Set: During the training process, for each training batch, we have access to the amplification or deamplification factor for each independent variable in the feature cascade. After traversing the entire dataset, we obtained three stacks containing the factor sets for all the images, where the three factors correspond to the independent variables of the three degrees of freedom in the feature cascade. To generalize to inference process, we have to store the factor set of each independent variable as a stack and compute the mathematical expectation of each independent variable. The set of factors used for inference is defined as follows:

$$\lambda_l^{\text{infer}} = E(\text{stack}(\lambda_{li})), (i = 1, 2, \ldots, m_l; l = 2, 3, 4) \tag{7}$$

where λ_l^{infer} is the lth factor corresponding to the lth independent variable for inference; λ_{li} is the ith factor of the lth independent variable in the training process; stack$(\cdot)$ denotes the factor stack corresponding to the independent variable; $E(\cdot)$ is the mathematical expectation operation; and m_l is the length of the lth $(l = 2, 3, 4)$ factor stack corresponding to the independent variable.

At this point, we can determine whether the factor of each independent variable is an amplification or deamplification factor, and retain all relevant variables corresponding to the amplification factor $(\lambda_i \geqslant 1)$.

5) Comprehensive Features of the Inference Stage: With the amplification factors and relevant variables in place, the next step is to generate the comprehensive features used for inference.

Step 9: Merging all the relevant variables (corresponds to $\lambda_i \geqslant 1$) outputs comprehensive features for the detection network in the inference stage. The comprehensive feature can be expressed as follows:

$$F^{\text{infer}} = \sum_i p(F_i)\lambda_i \mathbb{1}(\lambda_i \geqslant 1) \tag{8}$$

where $p()$ is the 1×1 conv projection and bilinear interpolation operation; $\mathbb{1}(\cdot) = 1$ if $\lambda_i \geqslant 1$, otherwise $\mathbb{1}(\cdot) = 0$.

6) Insights of KDN: As shown in Fig. 3, the KDN inputs three[7] "levels" of features from the feature extractor and outputs single-scale comprehensive features for the detection network. The three scale features with the corresponding amplification or deamplification factors determine the final feature output. In the inference procedure, we found that the KDN generates three equal amplification factors for different

[6]np.where() returns the indices of the elements that are nonzero.

[7]It depends on the independent degrees of freedom in the feature cascade.

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types of small object detection tasks. That is, the KDN outputs a set of amplification factors for the feature cascade.

This phenomenon means that although in the initial Faster R-CNN, except for the C_3 stage in the feature extractor, the other stages are not directly involved in the detection task, each stage produces a different level of features that are both independent of and complementary to each other. That is, we preconstructed a set of independent variables in the feature extractor can implement the renormalization group method. Finally, these variables are verified to be independent and complementary to each other. In other words, each independent variable in the feature cascade acts synergistically on the objective detection task without overlap or interference. This observation for the single-branch detector is consistent with the idea of "each in its own way" in the multibranch detectors' feature extractor represented by the FPN. We conclude that each deep network used as a feature extractor produces multilevel features, which certainly have different functions and can act synergistically in the detection task.

This insight satisfies the first principle feature "scaling" of the feature cascade mentioned in Section III-A. For example, averaging the F_{L_1} information of an object produces a factor that is very similar to the factor produced by averaging the F_{L_2} information of the object. This further validates our successful application of the renormalization group theory to the comprehensive FE problem.

C. RC Methods

In the following, we will discuss the RC methods. First, we will introduce the formulation of the RC method and then generalize it to a wider range of detection networks, such as multibranch detection networks using FPN series as feature extractors.

1) Formulation of RCs: We formulate the renormalization group method for FE as a RC. It is well known that any feature in a feature space can be represented by a set of feature bases and their corresponding coefficients, i.e., $f = \sum_i c_i b_i$, where c_i denotes the coefficient of the ith feature base; b_i denotes the ith feature base. By convention, in a feature pyramid extractor, different feature levels used for object detection are in different scale ranges but share weights in the head. We observed through KDN experiments with a single-branch detector that each of the preconstructed independent variables in the feature cascade is, indeed, independent and complementary to each other.

To ensure independence, we need a set of stricter bases to replace the initial independent variables in the feature cascade to cover the general conditions of feature pyramid extractors. To construct a set of independent bases, we first need to remove the possible correlation part between every two variables in the feature cascade. We formulate a feature cascade with three degrees of freedom as F_{L_1}, F_{L_2}, and F_{L_3}. F_{L_1} acts as the first independent variable and maintains the original variable type as F_{L_1}. Next, we take the second variable F_{L_2} minus the first independent variable's remainder $(F_{L_2} - F_{L_1})$ as the second independent variable. Finally, we keep the third variable F_{L_3} minus the first and second parts $(F_{L_3} - F_{L_2} - F_{L_1})$ as the third independent variable. Up to this point, three tighter

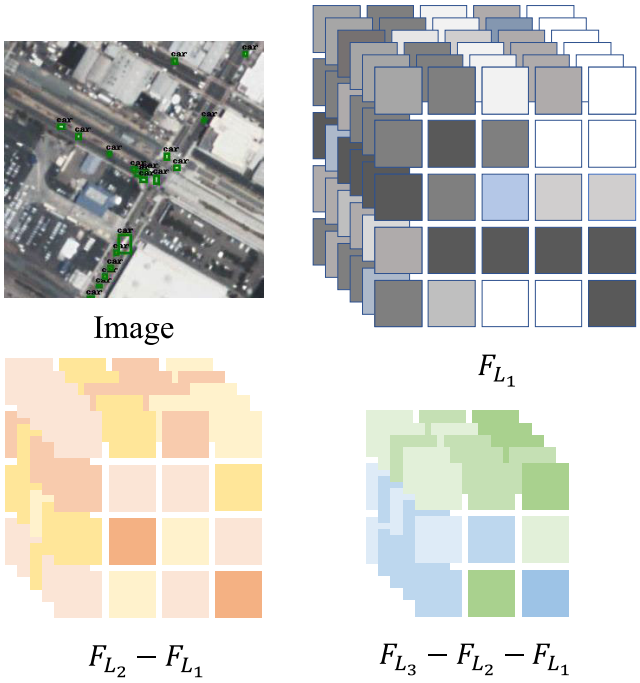


Fig. 4. Set of independent feature bases in a given feature space.

bases of independent variables are constructed, namely, F_{L_1}, $F_{L_2} - F_{L_1}$, and $F_{L_3} - F_{L_2} - F_{L_1}$, as illustrated in Fig. 4.

The RC is then established according to the conclusions of the renormalization group method of FE. In the case of single-branch detectors, we have a set of uniform amplification factors to form a comprehensive feature, i.e., $F = F_{L_1} + F_{L_2} + F_{L_3}$. We can use the Gaussian elimination method to calculate the connection strength of the set of uniform amplification factors. After that, we get $F_{L_1} + F_{L_2} + F_{L_3} = 4 \cdot F_{L_1} + 2 \cdot (F_{L_2} - F_{L_1}) + (F_{L_3} - F_{L_2} - F_{L_1})$. This connection strength ratio is 4:2:1, and we can generate one kind of RC called the 421 connection or 421C in short.

In this work, our focus is on difficult scale-preferred tasks, mainly the detection of small objects. While there are many possible sets of factors to construct a renormalized group for FE, we focus on adjusting the amplification factor of the first independent feature to renormalize the feature distribution of the feature extractor. The most direct connection strength comes from the observations of the KDN, forming a 421C. The calculation of the RC strengths for different amplification factors is straightforward, and it is sufficient to follow the same procedure that we used for the calculation of 421C.

Using 421 as a reference, if we have a deamplification factor of the first irrelevant variable F_{L_1}, the first connection strength 4 will be reduced. Conversely, if we have a larger amplification factor for the first relevant variable, the first connection strength 4 will be amplified. Under this condition, we can obtain a n21C, where $n > 4$ if the first independent variable corresponds to an amplification factor larger than the others; and $n < 4$ if the first irrelevant variable has a deamplification factor.

The simplest and most specific connection strengths are $\{4, 2, 1\}$. We use a random search method to find the appropriate connection strengths for each scale-preferred task. Also,

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n in n21C can be changed to adapt to different types of detection architectures. While there are multiple possibilities for connection strengths, as the experiments show, most design parameters are not particularly sensitive to exact values. Consistent with the scaling property of renormalization group theory in the feature cascade, we primarily use in this work the 421C that most typically satisfies this property. Next, we present the generalized RCs of multibranch detectors based on FPNs.

D. Insights of RCs

When the FPN-based multibranch detector is used directly to solve the scale-preferred problem, the detection branch with fewer positive samples generates a large number of interfering negative samples and background noise, which cannot be eliminated by using the focal loss function as shown in Fig. 1. These negative samples cannot be eliminated by using the focal loss function, because they are necessary samples and information generated during the training process according to the mechanism of "divide-and-conquer" and multisubtask parallel learning of different levels of features of the FPN, and they will only be generated continuously and will not be eliminated. However, in the case of scale-preferred, they seriously interfere with the training process and prevent the network from learning in the right direction.

Therefore, we designed the RC method to implement the mechanisms where multiscale/level features act synergistically and do not interfere with each other,[8] and focus on the dominant learning objective.[9]

On a single-branch detector, since there is only one detection branch, its mechanism for solving the main learning objective is naturally reached, and it is easy to realize the design idea of multiscale features acting in concert and focusing on the total learning objective using RCs.

However, on the multibranch detector, multiple detection branches are used for solving subtasks with different learning objectives. It is necessary to set up a pathway between the multilevel feature extractor and the multiple detection branches to guide the input features of multiple detection branches. It is no longer confined to the one-to-one correspondence between each detection branch and each level of features, so as to make the multiple detection branches truly realize the idea of "each in its own way" without interfering with each other, i.e., the "divide-and-conquer" design idea of FPN. Moreover, due to the introduction of RCs, the design idea of multiscale features acting together and focusing on the main learning objective is also inherited.

E. Generalized Applications of RCs

In the KDN experiments, the effectiveness of the RC was verified on single-branch detectors. In this section, we extend the RCs to multibranch detectors using the FPN family of feature extractors to test their validity. Sections III-E2–III-E4 describe methods for applying RCs to both single-branch and multibranch detectors, where the incorporation of RCs in multidetection-branch architectures can be subdivided into two approaches: economical and complete. Section III-F calculates the gradient of the RCs during backpropagation, proving the validity of n21Cs.

1) Overview: For the more difficult tasks of scale-preferred detection, tiny object detection, and small object detection in satellite imagery, we insert the RCs into 17 powerful mainstream detectors with different connection layers and different connection densities in the architecture. The process of applying RC is independent of the backbone, feature extractor, and head structure. Furthermore, to validate the performance of RCs on different scale-distributed tasks, we conducted extensive experiments on four typical satellite imager detection datasets with varying scale preferences, categories of interest, and ground resolution. The 17 representative detection or segmentation frameworks we use can be classified into two categories according to the number of branches in their detection networks, single-branch detectors and multibranch detectors. In the following, we describe the application of the RC to each of these two classes of detectors.

2) Applications on Single-Branch Detectors: The RCs can be used in single-branch detectors such as Faster R-CNN and region-based fully convolutional network (R-FCN), where the backbone network is the feature extractor. As shown in Fig. 3, KDN is a typical example of RC in a single-branch detector. This type of feature extractor can construct a set of independent variables as the feature cascade with three or four degrees of freedom, depending on the number of stages in the feature extractor. Since single-branch detectors have only one detection branch in the detection head, the RC is naturally inserted into the unique detection branch. In this article, we connect all the feature scales in the feature extractor to output comprehensive features of the whole image for the detection network. The RoI pooling layer in the detection network then extracts features from each object and feeds them into the head network for classification and regression tasks.

3) Economical Applications on Multibranch Detectors: Since the FPN series of feature extractors have multilevel features, there are various forms of constructing RCs on them. Depending on the task to determine on which or which scales of features in the feature extractor to insert the RC, we devised one of the simplest approaches that focus only on the first branch, the small object detection branch, as shown in Fig. 5. This simplest and most economical form uses only the first three levels of features $\{P_3, P_4, P_5\}$ as the feature cascade, in which each feature scale represents an independent variable. These independent variables with three degrees of freedom are fed into the RC operator. The RC outputs comprehensive features for the small object detection branch. Although the RC is only inserted into the small object detection branch, all features in the feature cascade are renormalized. That is, medium and large object detection branches are also affected by the RC operation. The right part of Fig. 5 shows an example

[8]Including the idea of "each in its own way," where each level feature addresses only the subtask for which it is responsible, without interfering with each other.

[9]Without generating a large number of negative samples from objects at other scales and background noise both interfering activations are equivalent to the design idea of fewer positive sample branches acting in concert with positive sample branches, focusing on the detection objective of the positive sample branches.

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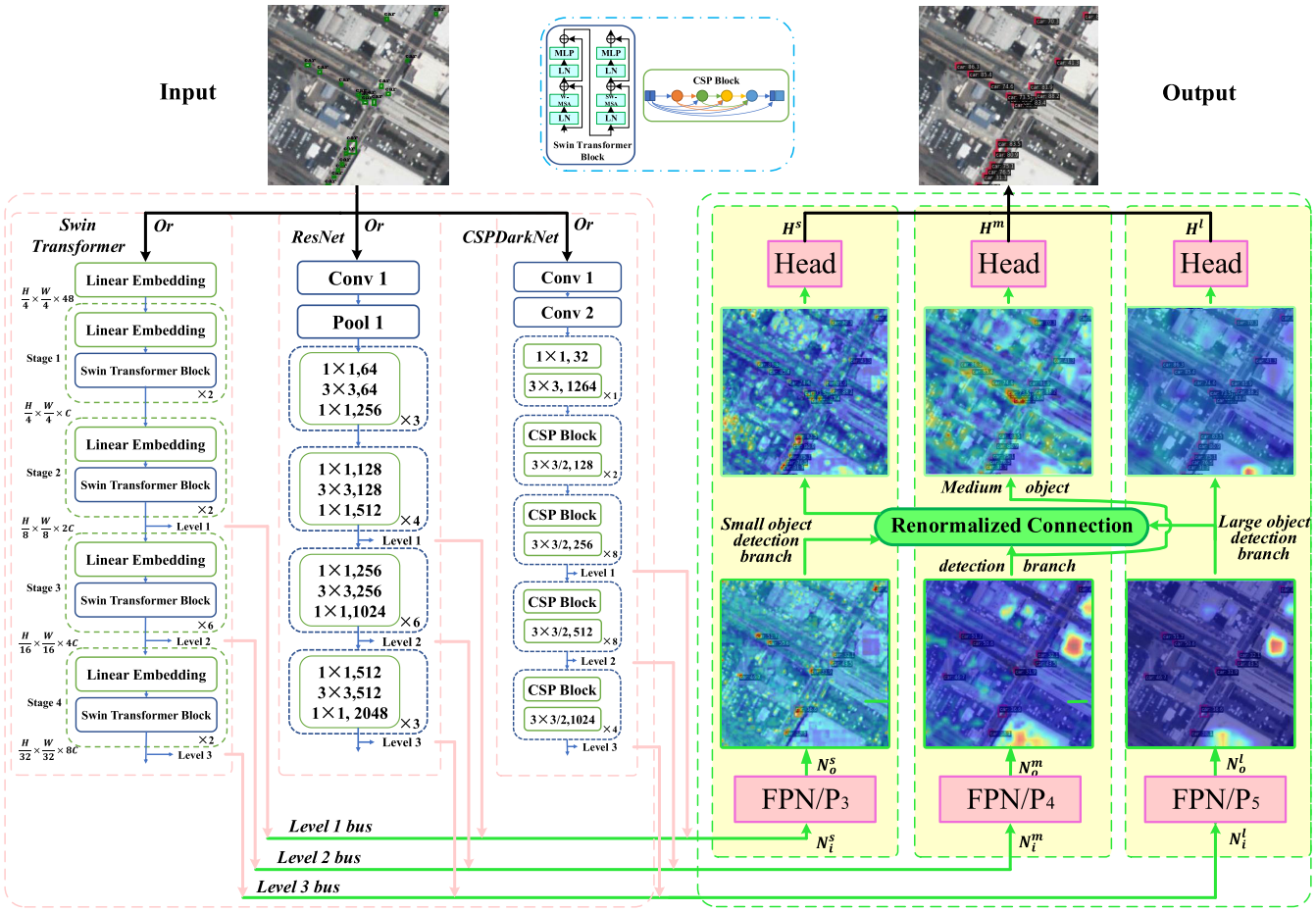


Fig. 5. Architecture of the detectors embedded with the RC (economical form). Note that the FPNs have different connection layers and densities as shown in Fig. 2(b). For simplicity, the fine structure of the feature extractor is not shown. Level 4 of Swin Transformer and ResNet are not shown for simplicity.

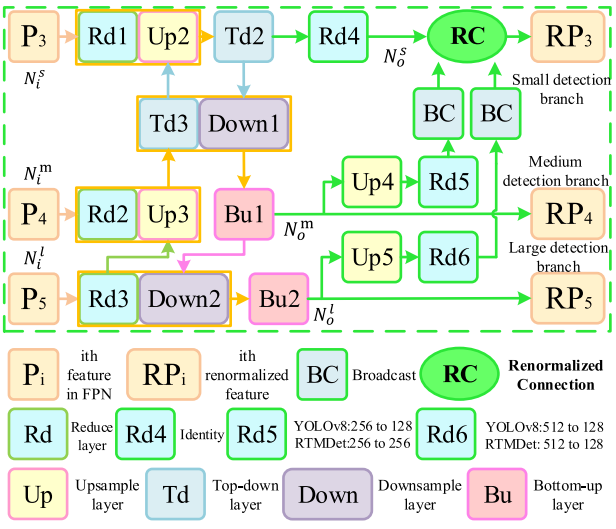


Fig. 6. Example structure of an economical RC in FPN-based multibranch detectors.

of input features (multilevel features of the FPN) and output features (renormalized features) of the economical RC.

We stabilize the connection strength as $\{4, 2, 1\}$ which satisfies the scaling invariance property of the feature cascade in the multilevel feature extractor to build a 421C at first. In our experiments, we can see that this simple and specific form of economical RC performs well in a variety of different scale-preferred tasks. Therefore, in this work, we focus on the simplest form of RC that satisfies the renormalization group theory and refer to this economic connection as E421C. We have equipped the E421C with multifarious multibranch detectors with different architectures.

In addition, we investigate different connection strengths to construct different types of RCs. To uniformly represent the different connection strengths of RCs, we use En21C, where $n \in R$. It is worth noting that the RC does not obey the scaling invariance property of the renormalization group when $n \neq 4$. Therefore, we mainly verify the effectiveness of E421C in our experiments. Other types of connections are also reported by randomly searching for connection strengths. For each RC with different connection strengths, we use a simplified connection strength ratio to determine n in En21C.

Apart from the effective E421C, we also investigated in detail the performance of the $\{5, 2, 1\}$ connection strength on different scale-preferred tasks. This type of connection is termed E521C. The doubling of the P_3 information in the input of E521C compared with E421C suggests that 521C is more focused on the small object detection branch than the 421C. Experiments show that E521C performs better or worse than

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E421C on different detectors (e.g., YOLOv7 and YOLOv8) and on different scale-preferred tasks. Figs. 5 and 6 illustrate the detection pipeline containing economical RCs. The first three levels in the feature extractor are rearranged to form the renormalized features and redirect the optimization during training.

The vanilla economical RC is a sparse linear connection that does not require additional learnable layers, nonlinear computations, and normalization operations. It minimizes modifications to the multibranch detector and ensures high inference speed. En21C is not only suitable for the most difficult scale-preferred task, i.e., purely tiny object detection task but also improves the accuracy of other scale-preferred tasks and scale-diversified tasks. In addition, we investigate connection methods that add three other representative deep learning techniques (1×1 conv, norm layer, and activation) on top of the linear En21C.

The applications in this subsection are based on the renormalization group theory, which was first applied to the tiny object detection task but has been well generalized to various scale-preferred tasks. Our goal is to pursue economy, efficiency, low density, and less computation in satellite image processing. The E421C achieves this goal and extends the application of renormalization group theory to feature renormalization. It can be used not only for difficult scale-preferred problems but also for scale-diversified datasets. We note that E421C is a linear connection method that does not increase the computational overhead of the detectors, yet helps the multilevel feature extractor to generate more independent and correct features, which is especially suitable for difficult scale-preferred tasks. Due to its robustness and generality, we mainly discuss the simplest linear E421C in this work. Designing better backbones or feature extractors as well as solving general object detection problems is not the focus of this work, so we opt for the simple design described above.

4) Complete Applications on Multibranch Networks: In the human brain, the density of interneuronal connections, i.e., synapses, is inversely proportional to memory capacity, and the combined effect of the two processes of strengthening important neural connections and degrading unimportant neural connections may lead to continuous optimization of the structure of the human memory system. Inspired by this brain neuroscience research, we have established a complete form of RC; that is, the use of all levels of features in the feature cascade to achieve an RC before the classification and regression tasks in each detection branch.

To validate the renormalization effect of the complete form, we experimentally compared the complete form of RC with the simplest economical form 421C mentioned in Section III-E3 on the extremely difficult scale-preferred task IPIU pure tiny object detection dataset. We find that the complete connection form reduces the mAP for small objects by a large margin (2.6% and 26.6% $\rightarrow$ 24.0%) and severely slows down inference (16.88 $\rightarrow$ 22.51 ms/f) when compared with the E421C. This result implies that the complete form is incapable of achieving the renormalized feature functions for difficult scale-preferred tasks. Therefore, to follow the mechanisms that strengthen important neural connections in the human brain and to adhere to the principle of Occam's razor, we will focus in this article on an economical approach to applying RCs to multibranch detectors.

The complete form of RCs has more connection strengths than the economical form. Since uniform connection strength does not further improve the performance on the basis of the simplest E421C, we only provide an adaptive feature pooling [9] inspired connection strength matrix as an alternative complete connection structure. The connection strength matrix is

$$C_{4\times4} = \begin{bmatrix} 1 & 0.15 & 0.1 & 0.1 \\ 1 & 0.3 & 0.3 & 0.25 \\ 1 & 0.25 & 0.25 & 0.25 \\ 0 & 0.3 & 0.35 & 0.4 \end{bmatrix}. \tag{9}$$

F. Gradients of RCs

The RC renormalizes the information flow in all phases of network learning, rearranging the feature flow in the forward propagation phase and reorienting the gradient flow in the backpropagation phase. The information flow in both phases together determines the learning direction of the network. Next, we analyze the gradient renormalization process in the backpropagation phase.

The essence of the SGD algorithm is that the gradient is large enough that the network stops learning when the gradient disappears. Typically, the gradient value has a very small value of around 0 and can easily disappear with training. RCs superimpose 3 or 4 (the independent degrees of freedom of the feature cascade) gradients on each detector branch. We need to consider three cases: 1) a unique detector branch on a single-branch detector; 2) a small object detection branch in multibranch detectors (economical form); and 3) any of the arbitrary detection branches in multibranch detectors (complete form), as shown in Fig. 7.

We know that the deeper layers have larger gradient values due to being closer to the classification layer. As a result, the gradient values of the detection branch output from the RC are larger than those of the original detection branch, which leads to greater parameter tuning, more complete network training, and longer effective training time. Most importantly, RCs can adjust the learning direction of the network to make it learn more efficiently, as we will observe in the experimental results with the AP curve. For example, Fig. 7(a) shows the gradient flow of the economical RC, where the blue arrows indicate the original gradient flow and the pink arrows indicate the gradient flow after adding the RC. The renormalized gradient of the small object detection branch can be expressed as follows:

$$\begin{aligned} &\mathrm{Grad}(w_s) \\ &= \mathrm{Grad}_{\mathrm{orig}} + \mathrm{Grad}_{n21C} \end{aligned} \tag{10}$$

$$\begin{aligned} &\mathrm{Grad}_{\mathrm{orig}} \\ &= \frac{\partial L_l}{\partial P_4}\cdot\frac{\partial P_4}{\partial C_4}\cdot\frac{\partial C_4}{\partial C_2} + \frac{\partial L_m}{\partial P_3}\cdot\frac{\partial P_3}{\partial C_3}\cdot\frac{\partial C_3}{\partial C_2} + \frac{\partial L_s}{\partial P_2}\cdot\frac{\partial P_2}{\partial C_2} \end{aligned} \tag{11}$$

$$\begin{aligned} &\mathrm{Grad}_{n21C} \\ &= \lambda_4\frac{\partial L_s}{\partial P_4}\frac{\partial P_4}{\partial C_4}\frac{\partial C_4}{\partial C_2} + \lambda_3\frac{\partial L_s}{\partial P_3}\frac{\partial P_3}{\partial C_3}\frac{\partial C_3}{\partial C_2} + \lambda_2\frac{\partial L_s}{\partial P_2}\frac{\partial P_2}{\partial C_2}. \end{aligned} \tag{12}$$

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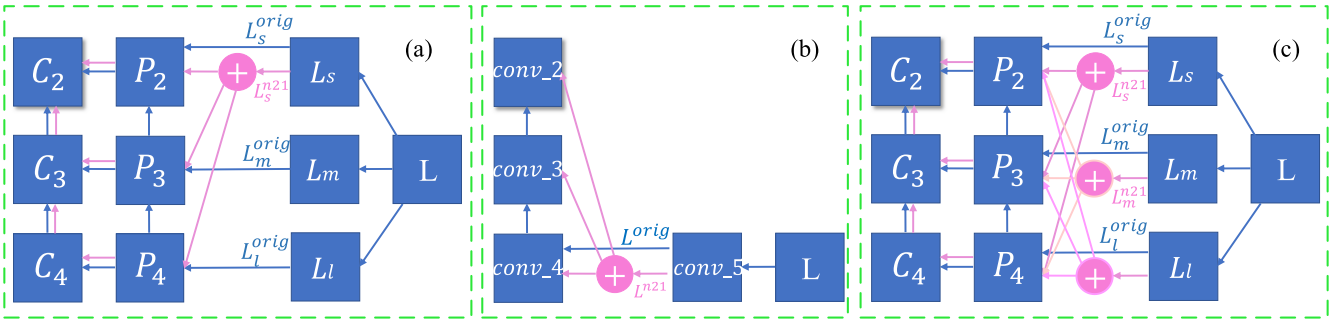


Fig. 7. Partial backpropagation computational graph of the three forms of RC. (a) Economical n21C on multibranch detectors. (b) RC on single-branch detectors (KDN). (c) Complete n21C on multibranch detectors.

In our experiments, we can see that the test accuracy can continue to be improved by continuing to train the fully convolutional one-stage object detector (FCOS) or Swin Transformer adding with the E421C. The gradient of the KDN is similar to E421C. The gradients of detectors equipped with complete RCs are also easier to derive, and we omit the formulaic descriptions of the latter two forms.

IV. Experiments and Discussion

RCs are available in the form of economical connections and complete connections. When used for object detection tasks, RCs can be added to different detection branches according to the task requirements. In order to RC whether RCs can improve the accuracy of different types of scale-preferred tasks, we provide experimental results on five representative datasets in this section.

A. Experimental Settings

1) Datasets: We conducted experiments on five different types of datasets that represent tasks with a diverse range of scale preferences, namely: 1) IPIU; 2) RSOD Aircraft [40]; 3) RSOD [40]; 4) NWPU VHR-10 [41]; and 5) MS COCO TOD-80 [12]. The first four satellite image datasets cover the following scenarios: 1) preferences for different scales, i.e., tiny objects, small objects, or objects of various scales; 2) number of categories ranging from 1 to 10; 3) purely tiny objects or objects of various sizes with large size variations; 4) densely or sparsely distributed objects; 5) used for research or applications; and 6) varying degrees of difficulty in detection. The fifth dataset is a natural scene 80-class small object detection dataset. We primarily use MS COCO TOD to train KDN and also to provide a broader perspective for evaluating RC methods.

IPIU: It is a remote sensing video dataset of purely high-density tiny objects. We selected a variety of scenes as detection images and manually annotated all moving vehicles, thus creating a difficult scale-preferred task for this work. The videos were acquired by Jilin-1 HD Dynamic Video Satellite with a ground resolution of 0.91 m/pixel and a frame rate of 10 frames/s. The carefully hand-annotated training and test set were acquired over San Diego Military Harbor, USA, in 2017. We provide the object detection benchmarks in this article. The scenes in this dataset are filled with a wide variety of man-made entities such as roads, overpasses, bridges, ships, containers, buildings, vehicles, cars, plants, and airplanes. The smallest and morphologically diverse entity "vehicle" was chosen for this challenge for detection or tracking. 95% of the objects are between 5 × 8 and 10 × 15 pixels in size. IPIU Dataset is a typical small object detection challenge with all the difficulties mentioned in Section I.

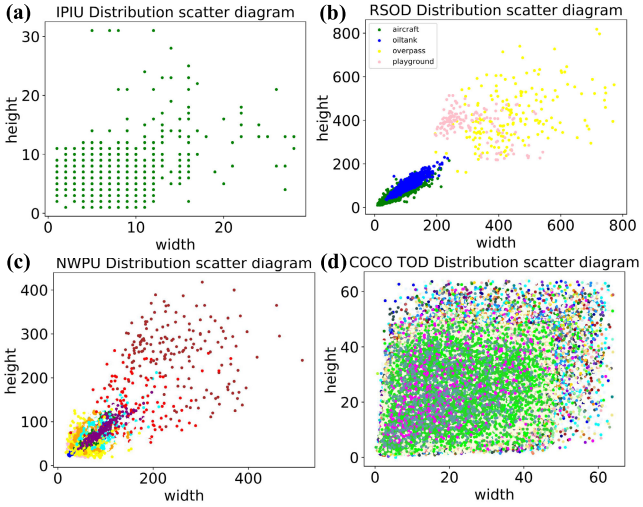


Fig. 8. Scatter plot of (width and height) points for all instances in the five datasets. (a) IPIU. (b) RSOD and RSOD Aircraft with green points. (c) NWPU VHR-10. (d) MS COCO TOD-80.

RSOD Aircraft: It has densely packed small "aircraft" instances with an image resolution of 0.5–2 m. It has 153 images containing small objects (whose area is less than 32 × 32 as defined in [42]), accounting for 34.3% of the total 446 images. The total number of instances is 5374, and the number of small instances is 1026 or 19.1%. In Fig. 8(b) (in green), the size of the instance is densely distributed in (0, 100). This dataset is a suitable representative of the scale-preferred task for small and medium-sized object detection.

RSOD: It is a four-category remote sensing benchmark where the size of the object instances varies a lot. As shown in Fig. 8(b), there is a large range of size variations for the four categories (aircraft, overpass, oil tank, and playground). The "overpass" and "playground" instances have a large range of sizes and are much larger than the "aircraft" and "oil tank" instances. This will visibly divide the four categories into

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two size variation tendencies. The split of the training and validation set is set to 5/1 as specified in [40]. The RSOD dataset is a task with diverse scale distributions.

NWPU VHR-10: It is a ten-category remote sensing object detection benchmark. It contains a total of 800 very high-resolution (VHR) remote sensing images. These images were extracted from Google Earth and the Vaihingen dataset and then manually annotated by experts. This dataset represents a task with diverse scale distributions, i.e., a scale-diversified problem.

MS COCO TOD-80: MS COCO Tiny Object Dataset is a generic 80-category natural scene benchmark. There are 54 196 images containing 276 702 instances. The training set contains 52 032 images and 265 667 annotated objects. The validation set contains 2164 images and 11 035 objects. There are a large number of varied scenes providing various contexts for 80 types of objects with different shapes. This dataset is used to learn the function of each scale feature in the feature extractor of single-branch detectors. Due to the complexity and diversity of its scenarios and objects of interest, TOD has a small dataset bias [43]. In addition, TOD is a typical scale-preferred problem, focusing only on multiple classes of small objects. In summary, this dataset is used to train the KDN and to validate the RC method.

2) Implementation Details: We perform a series of experiments on 21 well-designed detectors and embed the RC in 17 representative baselines for comparison.

For Faster R-CNN, R-fcn, and SSD-FPN, all experiments are performed on the TensorFlow framework and Object Detection API [44]. The images are resized to 900 × 600 without any data augmentation. The initial learning rate (lr) is set to 0.0003, reduced by a factor of 10 at 900K and 1.2M, and stopped at 1.5M.

For the remaining 18 detectors, all of the experiments are carefully implemented using the Pytorch library based on the MMdetection benchmark [45]. The input images are resized by a random flip operation. A step-decay lr strategy is adopted with an initial lr of 0.02; the number of linear warm-up steps is 500 with a warm-up ratio of 0.001; the momentum and weight decay are 0.9 and 0.0001, respectively. Schedule_1x: Training starts with an lr of 0.02 (0.002 for YOLOF) reduced by a factor of 10 at the 8th and 11th epochs, and stops at the 12th epoch. Unless otherwise stated, all networks are trained for a total of 12 epochs. Depending on the number of network parameters, each GPU samples $\{1, 2, 4, 8\}$ images per iteration. For YOLOv8: The initial lr is set to 0.01. The momentum and the weight decay are set to 0.937 and 0.0005, respectively. Also, the training is stopped after 500 epochs.

3) Evaluation Metrics: All experiments use COCO-style AP and AR [42] to evaluate the performance of unseen images. The total inference time and the inference time per frame for all experiments are measured on an NVIDIA GeForce 1080 GPU.

B. Results and Discussion

1) Results on IPIU: The performance comparisons between the detectors embedded with the proposed economic RC and the state-of-the-art approaches on the IPIU test set are shown in Table I. For simplicity, we use 421C to represent the economic RC that satisfies the scaling invariance property. In the complete connection experiments, we denote the economical 421 connection and the complete 421 connection by E421C and C421C, respectively. We designed KDN to generate and implement RC on single-branch detectors, and the amplification factors $(\lambda_2, \lambda_3, \lambda_4)$ of the KDN satisfy the ratio of 4:2:1. Therefore, we can use 421C to denote the RC of KDN on single-branch detectors. From the results, we can draw the following observations.

1) As shown in Table I and Fig. 1(a), the simplest form of RC performs well in the difficult scale-preferred task of tiny moving vehicle detection and drastically reduces inference activations of irrelevant feature branches in FPN. The economical RC acting with a focal loss would further eliminate negative samples and focus more on the hard negative samples generated by P_3. In other words, the RC implements the "divide-and-conquer" idea of FPN not only for information flow renormalization and learning redirection but also for the hard scale-preferred task.
2) In the first row of Table I, the mAP of Faster R-CNN-421C (KDN) reaches 26.3%, which exceeds a large fraction of powerful multibranch detectors. This result verifies that KDN is an efficient and effective feature renormalization method, realizing the application of renormalization group theory in FE.
3) By carefully observing the experimental results, anchor-based and anchor-free methods perform differently in detecting tiny objects. The best performer among the anchor-based methods is YOLOv7 with an $AP^{0.5:0.95}$ of 22.1%. RetinaNet achieves $AP^{0.5:0.95}$ of 18.1% after schedule_1x training, while FCOS achieves AP0.5 : 0.95 of 26.0%. With further training, the AP of FCOS improves significantly, reaching 35.5% after 120 epochs. The performance of Swin Transformer is at the same level as cascade R-CNN. Compared with Swin Transformer-421C, the AP of HTC-421C is improved by 1.8%. The powerful YOLOv8 achieves the highest AP of 55.1%, benefiting from the powerful PAFPN and anchor-free algorithms. The 421C improves the $AP^{0.5:0.95}$ of YOLOv8 by another 0.4%.
4) The inference time for the IPIU test set also varies with the network structure. Cascaded refinement increases the inference time, as in the case of cascade R-CNN and HTC baselines. The inference times for FCOS and RetinaNet are similar, both being 43–45 ms/frame. Swin Transformer's mask branch slows down the prediction. YOLOv8-421C is the fastest, at 16.89 ms/frame.
5) The highest $AP^{0.5:0.95}$ and AP^S for baselines and 421C on nine detectors are plotted in Fig. 9. We can see that 421C consistently achieves higher accuracy on all baselines. This proves that 421C does improve the accuracy of different types of high-performance detectors.
6) As shown in Table II, we can see that the complete form degrades the performance in terms of both mAP and inference speed with the addition of RCs to all branches. This further demonstrates that the economical form of

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TABLE I

AP AND AR OF DETECTORS WITH RC AND BASELINES ON IPIU TEST SET. THE IMAGES ARE RESIZED TO 600 × 600. YOLOV8 REQUIRES 640 × 640 RESIZED IMAGES AS INPUT AND ADOPTS MULTISCALE TESTING. T: TOTAL INFERENCE TIME (S). T/F: INFERENCE TIME PER FRAME (MS). THE BEST AP, AR, AND T ARE IN BOLD

Method	$AP^{0.5:0.95}$	$AP^{0.5}$	$AP^{0.75}$	AP^S	AR_1	AR_{10}	AR_{100}	AR^S_{100}	T(s)↓	T/f (ms) ↓
Faster R-CNN-421C (KDN)	26.3	75.3	9.5	26.3	2.3	18.0	40.1	40.1		
Cascade R-CNN [46]	13.4	31.5	9.3	13.4	16.5	16.5	16.5	16.5	43s	80.68
Cascade R-CNN-421C	**14.3**	**32.0**	**10.4**	**14.3**	**17.1**	**17.1**	**17.1**	**17.1**	44s	82.55
HTC [47]	14.7	35.5	**9.4**	14.7	18.9	18.9	18.9	18.9	65s	121.95
HTC-421C	**15.8**	**39.7**	9.0	**15.8**	**21.0**	**21.0**	**21.0**	**21.0**	73s	136.96
Libra R-CNN [48]	11.4	29.2	6.5	11.5	16.7	16.7	16.7	16.7	30s	56.29
Libra R-CNN-421C	**12.1**	**30.4**	**7.1**	**12.2**	**17.2**	**17.2**	**17.2**	**17.2**	31s	58.16
FCOS [49]	26.0	73.4	11.0	26.0	38.1	38.1	38.1	38.1	23s	43.15
FCOS-421C	**26.3**	**73.6**	**11.3**	**26.3**	**38.6**	**38.6**	**38.6**	**38.6**	24s	45.02
FCOS-421C(2x)	30.4	77.8	16.6	30.4	41.9	41.9	41.9	41.9	23s	43.15
FCOS-421C(10x)	**35.5**	**81.4**	**24.7**	**35.5**	**47.3**	**47.3**	**47.3**	**47.3**	**23s**	**43.15**
RetinaNet [21]	18.1	57.0	5.5	18.1	36.9	36.9	36.9	36.9	24s	45.02
RetinaNet-421C	**18.6**	**57.0**	**6.2**	**18.6**	**37.1**	**37.1**	**37.1**	**37.1**	**23s**	**43.15**
Swin Transformer [50]	13.8	35.3	8.2	13.9	18.6	18.6	18.6	18.6	41s	76.92
Swin Transformer-421C	**14.0**	**35.4**	**8.5**	**14.0**	**19.0**	**19.0**	**19.0**	**19.0**	43s	80.68
Swin Transformer-421C(10x)	**17.5**	**38.2**	**13.5**	**17.5**	**21.1**	**21.1**	**21.1**	**21.1**	**41s**	**76.92**
YOLOv7 [51]	22.1	68.4	6.2	22.1	2.0	15.6	36.6	36.6	9s	16.88
YOLOv7-421C	**26.6**	**75.5**	**10.1**	**26.6**	**2.3**	**18.1**	**40.2**	**40.2**	**9s**	**16.88**
RTMDet [25]	34.6	81.7	22.1	34.6	3.0	22.0	46.6	46.6	11s	20.63
RTMDet-421C	**35.5**	**83.9**	**22.4**	**35.5**	**3.0**	**22.3**	**47.7**	**47.7**	**11s**	**20.63**
YOLOv8 [20]	55.1	93.9	57.1	55.1	**3.7**	30.5	63.7	63.7	10s	18.76
YOLOv8-421C	**55.5**	**94.2**	**57.6**	**55.5**	3.6	**30.9**	**64.4**	**64.4**	**9s**	**16.89**

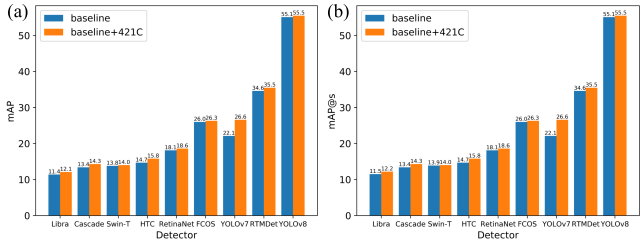


Fig. 9. (a) $AP^{0.5:0.95}$ (%) and (b) AP^S (%) comparison between 421C embedded detectors (orange) and baselines (blue) on IPIU test set.

TABLE II

AP AND AR OF YOLOV7 WITH ECONOMICAL AND COMPLETE RC METHODS: C421C AND E421C ON IPIU TEST SET

Method	$AP^{0.5:0.95}$	$AP^{0.5}$	$AP^{0.75}$	AP^S	T(s) ↓
C421C	24.0	71.8	7.3	24.0	12s
E421C	**26.6**	**75.5**	**10.1**	**26.6**	**9s**
Method	AR_1	AR_{10}	AR_{100}	AR^S_{100}	T/f (ms) ↓
C421C	2.0	16.9	38.3	38.3	22.51
E421C	**2.3**	**18.1**	**40.2**	**40.2**	**16.88**

RC can achieve feature renormalization for FPN-based multibranch detectors in difficult scale-preferred tasks.

2) Results on RSOD Aircraft: The comparison of AP and AR of the RC and baselines on single-branch detectors (KDN) and multibranch detectors on RSOD Aircraft are summarized in Tables III and IV, respectively. The results of the complete 421 connection and a weighted complete connection method are listed in Table V.

1) As shown in Table III, KDN far outperforms the baseline on almost all criteria in two typical classical single-branch detectors (Faster R-CNN and R-fcn). The RC improves the $AP^{0.5:0.95}$ of Faster R-CNN by 9.3% on ResNet-50 and 4% on ResNet-101. Among all the R-fcn experiments, 421C has the most significant improvement in $AP^{0.5:0.95}$ with 11%. The RC is particularly effective for small object detection on the shallower backbone ResNet-50, exceeding the AP^S of Faster R-CNN by 9.8% (from 14.5% to 24.3%) and R-fcn by 9.2% (from 11.6% to 20.8%). There has been a consistent improvement in the accuracy of small and medium objects across all experiments. The results confirm that KDN achieves renormalization of features for single-branch detectors on small object detection tasks.
2) As shown in Fig. 1(b), the simplest form of RC, E421C, performs well in the scale-preferred task of small aircraft detection, drastically reducing the inference activations of irrelevant feature branch (P_5) in FPN. The economical RC with focal loss function will further eliminate negative samples and focus more on the hard negative samples generated by P_3. That is, in the RSOD Aircraft scale-preferred task, the RC implements

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TABLE III
AP AND AR OF SINGLE-BRANCH DETECTORS FASTER R-CNN AND R-FCN WITH RC THIS IS THE IMPLEMENTATION OF KDN, ALL BASELINES, AND SSD-FPN ON RSOD AIRCRAFT VAL SET. AP: $AP^{0.5:0.95}$

Method	Backbone	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}	T(ms) ↓
Faster R-CNN [52]	R50	47.3	87.6	45.2	14.5	51.4	**74.1**	5.9	40.8	57.4	25.2	61.5	**78.4**	29.9
Faster R-CNN-421C	R50	**56.6**	**89.5**	**66.4**	**24.3**	**61.1**	72.2	**6.1**	**43.9**	**62.6**	**32.3**	**67.3**	75.2	**28.7**
Faster R-CNN	R101	52.0	88.7	55.6	19.8	56.6	**75.3**	5.8	**43.5**	61.6	30.8	65.8	**79.9**	29.1
Faster R-CNN-421C	R101	**56.0**	**91.2**	**65.5**	**26.5**	**60.1**	69.2	**5.9**	43.4	**62.4**	**35.0**	**66.7**	72.8	**27.6**
R-fcn [53]	R50	41.4	81.2	35.9	11.6	44.5	64.4	5.4	36.5	49.4	18.7	53.1	**70.8**	28.7
R-fcn-421C	R50	**52.4**	**86.4**	**59.0**	**20.8**	**57.2**	**63.7**	**5.9**	**41.9**	**59.0**	**29.2**	**64.2**	66.3	**27.1**
R-fcn	R101	41.8	80.5	40.0	13.8	44.8	64.3	5.3	35.7	49.9	20.3	53.5	**70.8**	29.7
R-fcn-421C	R101	**52.8**	**86.8**	**60.6**	**21.2**	**57.5**	**65.8**	**5.9**	**42.3**	**60.0**	**33.3**	**64.3**	69.5	**29.3**
SSD-FPN [21]	R50	41.9	71.5	44.3	12.0	46.4	55.0	5.8	36.4	52.9	24.1	57.8	60.9	26.2

the "divide-and-conquer" idea of FPN and observably rearranges the subtask assignment.

3) Table IV lists the AP and AR results of 13 well-designed detectors with multibranch detectors inserted with 421C and 17 baselines, including six powerful R-CNN-based two-stage baselines, 3 instance segmentation baselines, three anchor-free one-stage baselines, two transformer baselines, one combined with RoI extractor, and three real-time detectors (including the state-of-the-art YOLOv8). With all robust baselines, all 421C experiments achieve performance gains on most of the evaluated metrics. The 421C does not add any additional trainable layers (convolution/multihead attention) but performs well on different types of detectors, demonstrating the importance of enriching features. Compared with cascaded R-CNN, 421C improves $AP^{0.5:0.95}$ by 0.9% and AP^S by 2.5%. Combining 421C with the generic RoI extractor (GRoIE) RoI extractor increases AP^S by 0.2% but decreases $AP^{0.5:0.95}$ by 1.3%. YOLOv8-421C achieves the highest AP of 71.5%, exceeding the powerful baseline by 0.5%.
4) In Section III-E4, we introduced the complete 421 connection method and the weighted complete connection method. As shown in Table V, C421C on Swin Transformer mask R-CNN ($AP^{0.5:0.95}$:64.5%, AP^S:42.7%, AP^M:66.6%, AP^L:76.7%) further improves the accuracy of E421C ($AP^{0.5:0.95}$:64.1%, AP^S:44.6%, AP^M:66.0%, AP^L:75.9%). While C421C has the same AP as E421C (64.1%), it drops AP^S by 1.4%. Weighted complete connection gets an AP^S of 43.2%, which is 0.5% higher than C421C, but 1.4% lower than E421C. Also, it has the same AP as E421C. All observations show that the simplest economical RC is the economical choice for small object detection.
5) From the "T" and "T/f" columns in Table IV, the fastest algorithm is YOLOv8. YOLOv8 and RTMDet enable real-time detection. Mask prediction slows down the inference process. Portable networks predict fast. Overall, the inference time of the detectors is acceptable.
6) Fig. 10 shows the AP results of the baseline, GRoIE, and 421C. During training, 421C has higher AP scores than GRoIE, with 81 scores higher than 63.0% out of a total of 120 results. Especially in AP^S, 421C has 36 scores higher than 40.0%, which is much higher than the other two methods.

3) Results on RSOD: The comparison of the performance of a series of n21C and the corresponding baseline on RSOD is reported in Table VI. The best results are shown in bold. The n21C has been inserted into cascade R-CNN, HTC, Swin Transformer, RTMDet, YOLOv7, and YOLOv8. Each baseline represents a typical structure in the CV task. From these results, we have the following observations.

1) As shown in Table VI, n21Cs outperform all strong baselines in terms of $AP^{0.5:0.95}$. The AP scores of n21C are competitive with a variety of well-designed networks, cascade R-CNN, Swin Transformer, and YOLOv8. Specifically, the $AP^{0.5:0.95}$ of 421C improves by 1.7%, 0.8%, and 0.3%, respectively, when compared with the best AP score of the three baselines mentioned above, proving the effectiveness of n21C across a variety of modern detectors. These validate the good renormalization effect of the economical n21C on the task with diverse scale distributions, a multiclass satellite object detection dataset.
2) Let us see how the 421C performs on small objects. As shown in the "AP^S" column, the 421C outperforms the powerful HTC by 2.2% in AP^S, 0.3% in AP^M, and 0.2% in AP^L. In AP^S, the 421C outperforms the Cascade R-CNN by 1.5%. The last three rows show the highest AP^S for YOLOv8-based PAFPN, 421C and 521C, with 421C achieving 0.7% improvement in AP^S, 1.1% in AP^M, 0.7% in AP^L, and 0.3% in AP; and 521C achieving the highest AP^S score of 52.6%.
3) The inference times are listed in the last two columns. YOLOv8-n21C achieves a real-time detection of 25.31 ms per frame. The 421C reduces the total inference time of HTC and Swin Transformer by 2 and 1 s, respectively. This suggests that n21C is a lightweight and fast extractor, with no time overhead on the baselines.
4) The $AP^{0.5:0.95}$ and AP^S statistics of YOLOv8 and YOLOv8-421C are plotted in Fig. 11. From the plots,

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TABLE IV

AP AND AR OF DETECTORS WITH 421C AND BASELINES ON RSOD AIRCRAFT VAL SET. IMAGE SIZE: 1333 × 800, EXCEPT FOR YOLOV8 1280 × 1280. BACKBONE: RESNET-50 FOR CONVNET DETECTORS, SWIN TRANSFORMER: SWIN-T, YOLOV8: CSPDARKNET, CENTERNET: RESNET-18, AND CORNERNET: HOURGLASSNET-104. 421 CONNECTION ADDED ON THE TOP–DOWN [421C (A)] AND BOTTOM–UP PATH [421C (B)] OF PAFPN

Method	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}	T (s) ↓	T/f (ms) ↓
Cascade R-CNN [46]	65.9	95.2	80.7	41.5	68.3	**78.4**	70.5	70.5	70.5	50.4	73.3	81.1	10s	131.57
Cascade R-CNN-421C	**66.8**	95.4	**84.4**	44.0	**69.5**	78.2	71.6	71.6	71.6	50.3	74.6	82.9	**10s**	131.57
Cascade+GRoIE [28]	66.0	**96.1**	81.8	44.1	68.3	75.3	71.1	71.1	71.1	53.1	73.8	78.9	22s	289.47
Cascade+GRoIE-421C	65.5	95.3	81.2	**44.2**	67.6	77.2	70.1	70.1	70.1	51.8	72.5	80.9	23s	302.63
Faster R-CNN [8]	63.0	95.2	76.5	35.8	65.7	**76.9**	68.3	68.3	68.3	47.5	71.0	81.4	6.6s	86.84
Faster R-CNN-421C	**64.2**	**95.4**	**79.6**	**43.6**	**66.6**	74.1	69.6	69.6	69.6	51.7	72.3	77.7	6.9s	90.78
Grid R-CNN [54]	66.1	95.2	81.1	43.0	68.5	**78.2**	70.6	70.6	70.6	50.4	73.4	82.0	8.4s	110.53
Grid R-CNN-421C	**66.4**	**95.9**	**82.4**	**44.5**	**68.4**	77.7	71.4	71.4	71.4	53.8	73.6	82.4	9s	118.42
HTC [47]	66.7	95.8	**83.5**	42.9	69.2	79.0	71.5	71.5	71.5	51.9	74.1	83.3	15s	197.36
HTC-421C	**67.6**	**95.8**	82.9	**43.0**	**69.8**	**80.5**	72.1	72.1	72.1	53.3	74.5	84.4	**15s**	197.36
Sparse R-CNN [55]	34.8	61.4	36.4	**15.5**	37.9	52.8	58.7	58.7	58.7	26.6	63.2	75.4	6s	78.94
Sparse R-CNN-421C	**36.8**	**62.5**	**42.2**	13.4	**40.8**	**54.6**	57.4	57.4	57.4	22.6	62.0	79.4	7s	92.10
Mask R-CNN [56]	64.7	95.1	**81.1**	41.8	67.2	**77.3**	69.7	69.7	69.7	50.0	72.4	81.0	9s	78.94
Mask R-CNN-421C	**65.3**	**95.1**	80.6	**42.9**	**67.7**	77.2	70.1	70.1	70.1	51.2	72.6	81.3	10s	131.57
PANet [9]	64.0	95.1	78.9	41.0	66.4	74.9	68.7	68.7	68.7	51.0	71.2	78.4	7s	92.10
PANet-421C(A)	64.7	**95.2**	**81.0**	**43.1**	67.1	**77.4**	70.1	70.1	70.1	50.8	72.6	81.4	**7s**	92.10
PANet-421C(B)	**65.0**	94.2	80.7	42.1	**67.5**	77.0	69.7	69.7	69.7	49.7	72.5	81.1	**7s**	92.10
Libra R-CNN [48]	64.1	**95.6**	77.7	39.2	66.8	77.2	69.3	69.3	69.3	51.0	71.6	81.1	7s	92.10
Libra R-CNN-421C	**65.4**	95.0	**81.7**	**43.6**	**67.5**	**77.7**	70.5	70.5	70.5	53.7	72.7	81.4	**7s**	92.10
FCOS [49]	51.7	91.2	52.6	23.1	56.1	61.3	58.8	58.8	58.8	35.1	62.7	66.1	6s	78.94
FCOS-421C	**54.3**	**91.6**	**60.3**	**23.2**	**59.2**	**65.5**	60.9	60.9	60.9	33.7	65.3	70.4	**6s**	78.94
Centernet [57]	46.7	93.1	38.9	17.7	48.4	66.7	51.8	51.8	51.8	28.4	54.3	70.6	3.1s	40.78
CornerNet [58]	48.4	68.0	58.5	17.6	64.7	56.1	68.8	68.8	68.8	46.5	71.6	83.9	72.4s	952.63
YOLOF [59]	45.5	82.9	45.5	9.9	50.5	63.5	54.7	54.7	54.7	17.7	60.3	69.9	4s	52.63
DETR [60]	56.1	90.7	65.3	19.2	60.8	68.8	64.4	64.4	64.4	32.7	69.4	76.2	6.4s	84.21
Swin Transformer [50]	63.6	**96.1**	78.2	42.6	65.8	74.7	68.2	68.2	68.2	50.1	70.7	77.8	12s	157.89
Swin Transformer-421C	**64.1**	95.9	**78.5**	**44.6**	**66.0**	**75.9**	69.2	69.2	69.2	53.0	71.2	80.0	12.4s	163.15
RTMDet [25]	71.5	97.4	**89.8**	51.4	73.1	**83.5**	6.6	50.7	75.6	60.3	77.5	86.8	2s	26.31
RTMDet-421C	**71.5**	**97.5**	89.4	**52.4**	**73.1**	82.7	6.6	50.8	75.7	60.3	77.6	86.1	**2s**	26.31
YOLOv7 [51]	66.4	97.1	81.2	44.5	**68.4**	79.0	6.6	**48.6**	**72.3**	**58.4**	73.8	**83.8**	1s	13.15
YOLOv7-421C	**66.5**	**97.2**	**82.0**	**45.4**	68.2	**79.0**	**6.6**	48.4	72.2	57.6	**73.9**	82.3	**1s**	13.15
YOLOv8 [20]	71.0	**97.1**	**88.1**	**52.3**	72.2	82.9	6.8	50.8	76.0	62.1	77.6	86.7	1s	13.15
YOLOv8-421C(A)	**71.5**	**97.1**	87.4	51.7	72.6	**84.6**	6.8	51.1	75.8	61.1	77.4	87.3	**1s**	13.15
YOLOv8-421C(B)	**71.5**	**97.1**	86.3	52.1	**73.1**	82.6	6.7	51.0	75.9	61.6	77.5	86.3	**1s**	13.15

it can be seen that 421C consistently maintains high accuracy during the training process and achieves the highest AP. This proves that the 421C can continuously improve its performance on tasks with diverse scale distributions.

4) Results on MS COCO TOD-80: Table VII lists the performance of KDN (also known as 421C) on a single-branch detector compared with FPN-based multibranch detectors and an NAS method on the MS COCO TOD-80 dataset. Table VIII shows the AP and AR results of Cascade R-CNN-421C, YOLOv8-421C, and Swin Transformer-421C. From these results, we have the following observations.

1) In Table VII, we can observe that adding a RC to the single-branch detector Faster R-CNN (an implementation of KDN) greatly improves the performance of the 80-class small object detection task. KDN outperforms the listed multibranch detectors. NASNet, on the other hand, was not able to converge in an acceptable time because even one step of training took a long time. This verifies that RC can cope with the multiclass scale-preferred problem.

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TABLE V
AP AND AR OF SWIN TRANSFORMER MASK R-CNN WITH TWO TYPES OF COMPLETE CONNECTION METHODS: C421C AND WEIGHTED COMPLETE CONNECTION (WEIGHTEDCC) ON RSOD AIRCRAFT VAL SET. BACKBONE: SWIN-T

Extractor	$AP^{0.5:0.95}$	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L
C421C	**64.5**	96.0	**80.1**	42.7	**66.6**	**76.7**
WeightedCC	64.1	**96.8**	78.1	**43.2**	66.2	76.6
Extractor	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}
C421C	69.0	69.0	69.0	50.8	71.4	80.1
WeightedCC	69.1	69.1	69.1	52.0	71.3	80.8

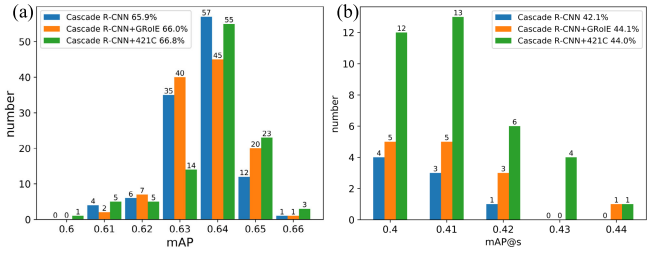


Fig. 10. RSOD Aircraft Val set inference results of (a) $AP^{0.5:0.95}$ and (b) AP^S. Statistical bar plots of Cascade R-CNN (blue), cascade R-CNN + GRoIE (orange), and cascade R-CNN-421C (green). After each training epoch, the three detectors are evaluated on the whole Val set to obtain results. Each interval contains an incremental AP of 0.01. For example, the first green bar in (a) means that there is one $AP^{0.5:0.95}$ result that falls in [0.60, 0.61].

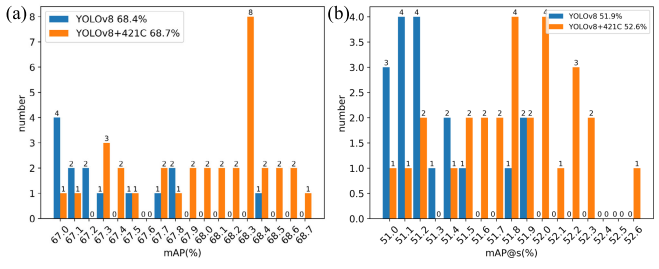


Fig. 11. RSOD validation set inference results of (a) $AP^{0.5:0.95}$ and (b) AP^S. Statistical bar plots of YOLOv8 (blue) and YOLOv8-421C (orange). After each training epoch, the two networks are evaluated on the entire Val set to obtain the results. Each label in the x-axis represents an AP value in %.

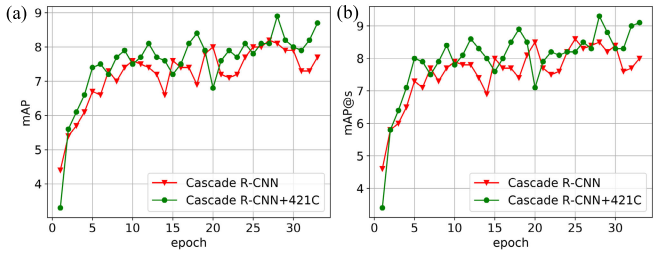


Fig. 12. MS COCO TOD-80 Val set inferential results of (a) $AP^{0.5:0.95}$ and (b) AP^S plots of cascade R-CNN-421C and baseline.

2) As shown in Table VIII, YOLOv8-421C achieves the highest AP of 11.4% and AP^S of 12.3%. cascade R-CNN-421C (8.9%) beats the baseline (8.2%) by 0.7% at $AP^{0.5:0.95}$. It achieves AP^S of 9.3% with 0.9% improvement compared to the baseline of 8.4%, proving that 421C is effective for small object detection under complex scenes.
3) Fig. 12 shows the AP of cascade R-CNN over the entire validation set (mainly considering $AP^{0.5:0.95}$ and AP^S) after each of the 33 epochs of training. The green line (421C) is almost always higher than the red line (baseline), suggesting that 421C brings a stable structured validity to the baseline. On multiclass complex natural scene scale-preferred datasets, 421C can effectively improve the prediction accuracy of small objects.

5) Results on NWPU VHR-10: The AP and AR comparisons of two typical n21C and baselines on NWPU VHR-10 are reported in Table IX. The 421C further improves the performance of two real-time detectors. YOLOv7-421C outperforms YOLOv7 by 0.5% in $AP^{0.5:0.95}$ and by 1.8% in AP^S. YOLOv8-421C improves by 0.4% in $AP^{0.5:0.95}$ and by 4.3% in AP^S. During training, the highest AP^S for YOLOv8 is 27%, while YOLOv8-421C reaches 33.5%. The 521C performs well on YOLOv7 but poorly on YOLOv8, suggesting that the doubled information obtained from the P_3 level in the scale-diversified task affects the performance of the anchor-based and anchor-free detectors differently. These results indicate that E421C performs well on the more complex multiclass arbitrary-sized satellite object detection dataset (the task with the most diverse scale distributions in this work).

6) Ablation Study: To gain a deeper insight into RCs, we continue to perform ablation experiments on satellite image datasets. First, we conduct a series of RC experiments on YOLOv7 with different connection strengths to investigate the renormalization effect of the uniform form 421C and other connections, as shown in Tables X–XII and Table IX, respectively. Second, we add three representative modules to the linear n21C and show their performance separately. Third, we compare two variants of E421C that take as input some of the features in the feature cascade: 1) $421C^{s\&m}$ and 2) $421C^{s\&l}$.

Ablation Study 1: In KDN, we use the amplification factor λ to describe the renormalization rate of all features in the feature cascade. Similarly, in the generalized RC n21C used in multibranch detector, we use connection strength to represent the same description of the amplification factor. In the scale-preferred task, we only consider adjusting the connection strength at the P_3 level.

1) As shown in Table X, we examined six amplified or deamplified factors compared to the 421 factors in the difficult scale-preferred tiny object detection task. The uniform connection strength of RC (421C) is superior to that of the other types of RCs. This study demonstrates that the simplest E421C that satisfies the scaling property of renormalization group theory is suitable for solving hard scale-preferred problems as described in Sections III-A and III-B6.
2) In the RSOD Aircraft experiments as in Table XI, we compare the performance of two typical n21Cs, the 421C, and the 521C, on a small object detection task (with large scale range compared to tiny objects). The YOLOv7-521C outperforms the 421C on mAP but performs poorly on AP^S and most ARs. However, the 521C

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TABLE VI

AP AND AR OF DETECTORS WITH n21C AND BASELINES ON RSOD VAL SET. IMAGE SIZE: 1333 × 800. YOLOV8: 1280 × 1280. BACKBONE: RESNET-50 FOR CONVNET DETECTORS, SWIN-T FOR SWIN TRANSFORMER, AND YOLOV8CSPDARKNET FOR YOLOV8. AP: $AP^{0.5:0.95}$

Method	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}	T(s) ↓	T/f (ms) ↓
Cascade R-CNN [46]	62.9	93.3	71.3	40.6	70.3	66.6	68.6	68.6	68.6	50.9	76.0	71.8	18s	113.92
Cascade R-CNN-421C	64.6	92.8	75.1	42.1	70.3	68.8	70.2	70.2	70.2	49.0	75.7	73.8	19s	120.25
HTC [47]	64.2	92.6	73.6	40.9	71.3	67.6	70.4	70.4	70.4	50.1	76.6	73.5	37s	234.18
HTC-421C	64.2	92.0	72.9	43.1	71.6	67.8	71.4	71.4	71.4	52.6	77.3	74.7	35s	221.52
Swin Transformer [50]	65.3	95.8	73.4	41.8	70.1	68.7	70.2	70.2	70.2	49.7	75.1	73.3	23s	145.57
Swin Transformer-421C	66.1	95.8	72.3	41.1	70.1	70.0	70.7	70.7	70.7	49.0	75.3	74.1	22s	139.24
RTMDet [25]	59.4	86.7	67.5	49.0	71.0	63.1	28.3	58.4	69.5	58.3	78.2	72.6	7s	44.3
RTMDet-421C	59.7	85.9	69.3	47.1	71.4	63.4	27.9	59.0	70.8	56.3	78.8	73.8	6s	37.9
YOLOv7 [51]	62.7	90.4	70.9	50.7	71.8	66.3	31.2	60.5	70.8	61.1	79.0	73.7	4s	25.31
YOLOv7-421C	62.9	91.4	72.1	48.6	71.8	66.4	32.3	60.7	71.4	59.5	78.8	74.0	3s	18.98
YOLOv7-521C	63.0	90.7	71.3	49.6	72.6	66.5	32.7	60.7	71.4	59.9	78.2	74.3	3s	18.98
YOLOv8 [20]	*68.4*	93.1	77.9	50.1	72.5	71.7	36.0	65.1	75.7	60.1	78.5	78.6	4s	25.31
YOLOv8-421C	**68.7**	93.5	79.6	49.6	74.3	71.8	36.2	64.2	74.4	60.3	79.7	77.2	4s	25.31
YOLOv8-521C	68.7	93.6	76.5	50.4	73.9	72.4	35.6	64.8	75.2	60.3	80.0	78.4	4s	25.31
YOLOv8 TopAP^S [20]	67.0	92.1	76.7	*51.9*	73.8	70.0	34.6	64.4	75.4	*61.7*	79.4	78.0	4s	25.31
YOLOv8-421C TopAP^S	67.3	91.9	75.0	**52.6**	74.9	70.7	34.6	64.3	75.4	**62.4**	79.7	78.3	4s	25.31
YOLOv8-521C TopAP^S	68.7	92.5	77.4	51.8	74.9	72.3	35.6	65.6	76.4	61.0	79.8	79.7	4s	25.31

TABLE VII

AP OF KDN EXPERIMENTS ON MS COCO TOD VAL SET. AP: $AP^{0.5:0.95}$

Method	Extractor	Image size	Steps	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}
R-fcn [53]	ResNet101	512 × 300	1.2M	7.5e-2	0.22	4.3e-2	7.1e-2	0.21	7.5e-2	0.22	4.3e-2	7.1e-2	0.21
SSD-FPN [21]	ResNet50-FPN	640 × 640	1.2M	2.9e-2	1.1e-2	1.1e-2	3.1e-2	1.0e-2	2.9e-2	1.1e-2	1.1e-2	3.1e-2	1.0e-2
SSD-FPN [21]	ResNet101-FPN	640 × 640	1.1M	1.5e-7	4.4e-7	2.2e-8	7.9e-7	5.0e-8	1.3e-6	2.0e-5	3.1e-5	1.4e-5	1.9e-4
NASNet-A [61]	ResNet101-NAS	300 × 300	713	0	0	0	0	0	0	0	0	0	0
Faster R-CNN-FPN [8]	ResNet50-FPN	640 × 640	1M	1.2	2.2	1.1	1.1	1.9	3.9	8.2	9.3	8.9	13.4
Faster R-CNN [52]	ResNet101-C_3	512 × 300	1M	1.5	3.9	0.8	1.3	4.2	3.9	7.9	9.1	8.2	15.7
Faster R-CNN-421C	ResNet101-KDN	512 × 300	1M	**3.9**	**7.6**	**3.3**	**3.8**	**6.5**	**7.8**	**14.0**	**15.7**	**15.1**	**20.7**

TABLE VIII

AP AND AR OF DETECTORS WITH 421C AND BASELINES ON MS COCO TOD-80 VAL SET. IMAGE SIZE: CASCADE R-CNN: 1333 × 800; SWIN TRANSFORMER: 600 × 600; AND YOLOV8: 640 × 640. BACKBONE: RESNET-50, SWIN-T, AND CSPDARKNETRESNET-50

Method	Epoch	$AP^{0.5:0.95}$	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	T(s) ↓	T/f (ms) ↓
Cascade R-CNN [46]	33	8.2	13.7	8.8	8.4	**10.9**	17.9	17.9	17.9	17.8	26.2	258s	119.22
Cascade R-CNN-421C	33	**8.9**	**14.6**	**9.8**	**9.3**	10.7	**19.3**	**19.3**	**19.3**	**19.1**	30.8	262s	121.07
Swin Transformer-421C	12	8.0	15.5	8.0	8.7	10.6	18.7	18.7	18.7	18.8	31.2	180s	83.17
YOLOv8 [20]	500	11.3	19.1	11.7	12.2	13.5	15.7	29.3	33.7	33.7	46.8	46s	21.26
YOLOv8-421C	500	**11.4**	18.4	**12.3**	**12.3**	**15.2**	**16.0**	**29.6**	**34.2**	**33.7**	**48.0**	**46s**	**21.26**

TABLE IX

AP AND AR OF DETECTORS WITH 421C, 521C, AND BASELINES ON NWPU VHR-10 TEST SET. (ABLATION STUDY 1.4.) IMAGE SIZE: 1280 × 1280

Method	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}	T (s) ↓	T/f (ms) ↓
YOLOv7 [51]	46.6	85.0	45.4	18.0	44.1	50.1	20.6	50.1	57.9	31.3	57.0	55.9	3s	15.3
YOLOv7-421C	47.1	84.4	45.4	19.8	**45.6**	48.5	20.5	50.0	59.1	32.8	57.7	55.8	3s	15.3
YOLOv7-521C	**47.6**	**86.4**	**46.2**	**21.5**	44.9	**52.0**	**21.2**	**51.5**	**59.2**	**36.1**	**58.1**	**62.5**	3s	15.3
YOLOv8 [20]	61.2	92.8	67.7	18.3	59.5	59.9	25.0	61.6	68.5	37.3	67.9	65.5	4s	20.5
YOLOv8-421C	**61.6**	**93.2**	**68.9**	**22.6**	**61.1**	**60.9**	24.9	61.7	**68.5**	**35.5**	**69.4**	**69.7**	4s	20.5
YOLOv8-521C	61.1	92.9	68.5	14.7	60.5	60.1	**25.1**	**62.0**	68.4	33.8	68.1	64.0	4s	20.5

performed worse than 421C on YOLOv8. YOLOv7's results on detecting small-/medium-sized objects differ from the results on the difficult scale-preferred IPIU task.

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TABLE X
ABLATION STUDY 1.1. AP AND AR OF YOLOV7-BASED RC WITH DIFFERENT CONNECTION STRENGTHS ON IPIU TEST SET

Method	$AP^{0.5:0.95}$	$AP^{0.5}$	$AP^{0.75}$	AP^S	AR_1	AR_{10}	AR_{100}	AR^S_{100}	T(s)↓	T/f (ms) ↓
YOLOv7 [51]	22.1	68.4	6.2	22.1	2.0	15.6	36.6	36.6	9s	16.88
YOLOv7-421C	**28.7**	**78.5**	**12.0**	**28.7**	**2.5**	**19.3**	**41.6**	**41.6**	11s	20.63
YOLOv7-521C	27.7	77.0	11.0	27.7	2.3	18.6	41.6	41.6	9s	16.89
YOLOv7-721C	27.9	77.6	11.2	27.9	2.4	18.7	41.3	41.3	12s	22.51
YOLOv7-821C	26.1	74.8	9.7	26.1	2.3	18.0	40.0	40.0	11s	20.63
YOLOv7-(3.1)21C	22.7	68.2	7.1	22.7	2.1	16.2	36.9	36.9	12s	22.51
YOLOv7-(3.5)21C	28.2	77.9	11.6	28.2	2.5	19.0	41.4	41.4	12s	22.51

TABLE XI
ABLATION STUDY 1.2. AP AND AR OF DETECTORS WITH 421C, 521C, AND BASELINES ON RSOD AIRCRAFT VAL SET

Method	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}	T (s) ↓	T/f (ms) ↓
YOLOv7 [51]	66.4	97.1	81.2	44.5	**68.4**	79.0	6.6	**48.6**	**72.3**	**58.4**	73.8	**83.8**	1s	13.15
YOLOv7-421C	**66.5**	**97.2**	**82.0**	**45.4**	68.2	**79.0**	**6.6**	48.4	72.2	57.6	**73.9**	82.3	**1s**	13.15
YOLOv7-521C	65.1	96.9	78.6	41.6	67.5	77.0	6.1	47.7	71.2	55.4	73.2	81.1	**1s**	13.15
YOLOv8 [20]	71.0	**97.1**	**88.1**	**52.3**	72.2	82.9	**6.8**	50.8	**76.0**	**62.1**	**77.6**	86.7	1s	13.15
YOLOv8-421C(A)	**71.5**	**97.1**	87.4	51.7	72.6	**84.6**	**6.8**	**51.1**	75.8	61.1	77.4	**87.3**	**1s**	13.15
YOLOv8-421C(B)	**71.5**	**97.1**	86.3	52.1	**73.1**	82.6	6.7	51.0	75.9	61.6	77.5	86.3	**1s**	13.15
YOLOv8-521C	71.3	**97.1**	87.7	51.6	**73.1**	82.1	**6.8**	51.0	75.8	61.2	**77.6**	86.3	**1s**	13.15

TABLE XII
ABLATION STUDY 1.3. AP AND AR OF DETECTORS WITH 421C, 521C, AND BASELINES ON RSOD VAL SET

Method	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AP^M	AP^L	AR_1	AR_{10}	AR_{100}	AR^S_{100}	AR^M_{100}	AR^L_{100}	T(s) ↓	T/f (ms) ↓
YOLOv7 [51]	62.7	90.4	70.9	50.7	71.8	66.3	31.2	60.5	70.8	61.1	79.0	73.7	4s	25.31
YOLOv7-421C	62.9	91.4	72.1	48.6	71.8	66.4	32.3	60.7	71.4	59.5	78.8	74.0	3s	18.98
YOLOv7-521C	63.0	90.7	71.3	49.6	72.6	66.5	32.7	60.7	71.4	59.9	78.2	74.3	3s	18.98
YOLOv8 [20]	*68.4*	93.1	77.9	50.1	72.5	71.7	36.0	65.1	75.7	60.1	78.5	78.6	4s	25.31
YOLOv8-421C	**68.7**	93.5	79.6	49.6	74.3	71.8	36.2	64.2	74.4	60.3	79.7	77.2	4s	25.31
YOLOv8-521C	68.7	93.6	76.5	50.4	73.9	72.4	35.6	64.8	75.2	60.3	80.0	78.4	4s	25.31
YOLOv8 TopAP^S	67.0	92.1	76.7	*51.9*	73.8	70.0	34.6	64.4	75.4	*61.7*	79.4	78.0	4s	25.31
YOLOv8-421C TopAP^S	67.3	91.9	75.0	**52.6**	74.9	70.7	34.6	64.3	75.4	**62.4**	79.7	78.3	4s	25.31
YOLOv8-521C TopAP^S	68.7	92.5	77.4	51.8	74.9	72.3	35.6	65.6	76.4	61.0	79.8	79.7	4s	25.31

TABLE XIII
ABLATION STUDY 2. AP AND AR OF YOLOV7 BASED n21C ADDED WITH EXTRA REPRESENTATIVE MODULES ON IPIU TEST SET. 1 × 1 CONVOLUTIONAL LAYER, NORM OR BN, AND ACTIVATION: RELU. THE BEST AP, AR, AND T ARE IN BOLD

Method	1 × 1 conv	norm	act	$AP^{0.5:0.95}$	$AP^{0.5}$	$AP^{0.75}$	AP^S	AR_1	AR_{10}	AR_{100}	AR^S_{100}	T(s)↓	T/f (ms) ↓
YOLOv7 [51]				22.1	68.4	6.2	22.1	2.0	15.6	36.6	36.6	9s	16.88
YOLOv7-421C				**28.7**	**78.5**	12.0	**28.7**	**2.5**	**19.3**	41.6	41.6	11s	20.63
YOLOv7-521C				27.7	77.0	11.0	27.7	2.3	18.6	41.6	41.6	9s	16.89
YOLOv7-521C	✓			28.2	77.6	11.8	28.2	2.4	18.8	41.8	41.8	11s	20.63
YOLOv7-421C		✓		27.7	77.4	10.7	27.7	2.4	18.8	41.0	41.0	12s	22.51
YOLOv7-421C			✓	**28.7**	**78.5**	**12.2**	**28.7**	2.4	19.1	**42.0**	**42.0**	11s	20.63
YOLOv7-421C	✓	✓	✓	23.6	71.6	6.5	23.6	2.0	16.6	38.1	38.1	12s	22.51

3) As shown in Table XII, the performance of YOLOv8-521C is comparable to that of 421C, but with a higher AP for small objects. For YOLOv7, the performance of 521C is slightly better than that of 421C. This diverse scale-distributed dataset has similar performance for different types of RCs.

4) On the most diverse scale distributed NWPU dataset, 421C and 521C perform better on YOLOv8 and YOLOv7, respectively, as shown in Table IX. When performing YOLOv7-RC on scale-diversified datasets, the larger connection strength of P_3 can promote the AP^S, which is consistent with the general intuition

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Fig. 13. Visualized results of (a) YOLOv8-421C, (b) YOLOv8, and (c) GT on IPIU test set. Yellow boxes indicate false positives for YOLOv8. Orange boxes indicate missing detections for YOLOv8. The Wathet box indicates dark backgrounds but accurately detects results.

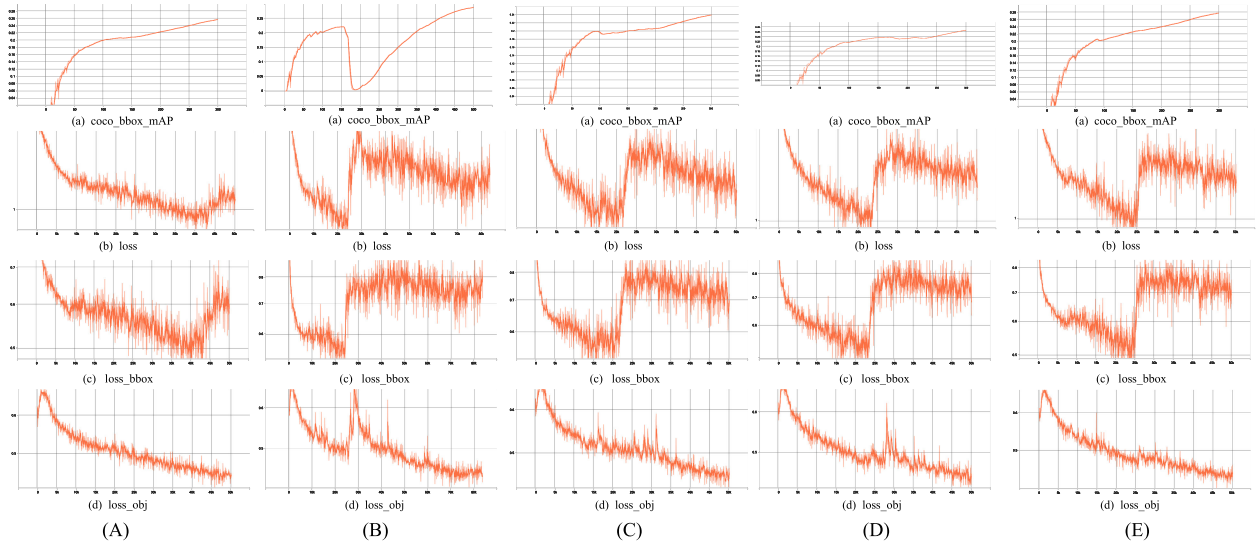


Fig. 14. AP and loss curves of YOLOv7 (baseline), RC, complete 421C, economical 421C, and economical 521C on IPIU Val set. (A) YOLOv7. (B) YOLOv7 + RC 500 epochs. (C) YOLOv7 + C421C. (D) YOLOv7 + E421C. (E) YOLOv7 + E521C. (a) Coco_bbox_mAP. (b) Loss. (c) Loss_bbox. (d) Loss_obj.

TABLE XIV
ABLATION STUDY 3. YOLOV8-421C AND ITS VARIANTS ON THE IPIU TEST

Method	AP	$AP^{0.5}$	$AP^{0.75}$	AP^S	AR_1	AR_{10}	AR_{100}	AR^S_{100}	T(s)↓
$421C^{s\&m}$	55.4	94.2	57.0	55.4	3.6	30.7	64.2	64.2	8s
$421C^{s\&l}$	55.4	94.1	57.2	55.4	3.6	30.6	64.3	64.3	9s
421C	**55.5**	**94.2**	**57.6**	**55.5**	**3.6**	**30.9**	**64.4**	**64.4**	9s

of weighted feature fusion. This further confirms that RC can bias the detector toward the objective of its embedded branch.

Ablation Study 2: We conducted a series of experiments on the IPIU test set by adding common modules used to enhance the representation capability on the linear n21C to investigate further potential factors affecting the performance gain. As shown in Table XIII, the performance of YOLOv7-421C with the added ReLU activation function is comparable to that of the initial linear 421C. The other modules do not improve the performance over the linear n21C but make it worse. This verifies that the linear RCs can handle the scale-preferred problem well, even under extreme conditions such as purely tiny object detection in satellite images.

Ablation Study 3: The 421C$^{s\&m}$ is a partial branch connected network where F^s and F^m pass through the 421 connection to enhance the output of 421C^s. Compared with 421C, the large object branch F^l does not participate in the detection of small objects in 421C$^{s\&m}$. Similarly, the medium object branch F^m is absent from the connection operation of 421C$^{s\&l}$. The mAP and AR results of 421C and its variants are given in Table XIV. From this table, we can see that the following hold.

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1) $421C^{s\&m}$ and $421C^{s\&l}$ have some performance improvements over the baseline, indicating that the 421 connection is useful for small object detection.
2) The 421C outperforms $421C^{s\&m}$ and $421C^{s\&l}$. These results demonstrate that both medium and large object branches' features are crucial for a complete description of the feature in the small object branch.
3) $421C^{s\&m}$ and $421C^{s\&l}$ achieve similar performance regardless of the choice of detection branch, suggesting that 421 connection is suitable for effective feature enhancement with fewer branches.

7) Analysis on Different Sizes: We will discuss how the proposed method affects the AP and AR at different object sizes. For full-size detection benchmarks, as shown in Fig. 11, 421C not only achieves higher mAP but also improves the AP for small objects. In the last two rows of Table VI, YOLOv8-421C increases the $AP^{0.5:0.95}$, AP^S, AP^M, and AP^L by 0.3%, 0.7%, 1.1%, and 0.7%, respectively; and increases the AR^S_{100}, AR^M_{100}, and AR^L_{100} by 0.7%, 0.3%, and 0.3%, respectively. In summary, 421C improves the detection performance of small, medium, and large objects.

8) Visualization of Predicted Results: To have some visualized observations, we further show the visualized results of YOLOv8-421C on the IPIU test set in Fig. 13. In Fig. 13, the three images in each column represent the results of YOLOv8-421C, YOLOv8, and GT on one image. We manually encircled the false detections, missing detections, and dark background detection results with yellow, orange, and Wathet boxes, respectively. Compared to YOLOv8, YOLOv8-421C predicts moving vehicles in complex scenes more accurately in the following aspects: 1) YOLOv8-421C has higher mAP and single instance AP than YOLOv8; 2) vehicles can be accurately detected even in dark backgrounds; 3) the 421C mitigates the problems of false detection and missing detection; and 4) it correctly identifies the desired objects and ignores other similar objects such as vehicles in the parking lots. From these visualization results, we can see that YOLOv8-421C can cope with the hard scale-preferred tasks in scenes that are blurred, low contrast, and full of interfering objects.

9) Comparison of AP and Loss Curves: In Fig. 14, we show the mAP and loss curves of the five YOLOv7-based detectors on the IPIU dataset. The left column shows the YOLOv7 baseline using PAFPN as the feature extractor. In the second column, we can see that the accuracy of PAFPN with RC decreases rapidly during training, but gains greater acceleration after the mAP value drops to 0 and reaches a higher value at the end of training. The loss curves of different types of $n21C$ have a similar trend, i.e., the total loss continues to decrease after a step change, while the baseline saturates at last. RC has the property of bottoming out quickly and recovering faster, unlike conventional deep network training. The phenomenon of RC (the second column) resembles the "hit bottom and rebound." It is able to rebound from a brief crisis.

The AP curves further demonstrate that the RCs do rearrange the multilevel features of the feature extractor during the training process and learn in a more correct direction, ultimately achieving better experimental results.

V. Conclusion

In this article, we studied the challenging problem of scale-preferred detection in satellite imagery, especially for the difficult tiny object detection. Inspired by the renormalization group theory, we design a KDN to implement the theory on efficient FE. According to the observations of KDN experiments, we abstract a class of RCs with different connection strengths called $n21C$.

After conducting experiments on scale-preferred tasks of the original YOLOv8-PAFPN, we observed that the "divide-and-conquer" design ideas of the FPN severely hampers detector learning due to the presence of a large number of negative samples at other scales as well as interfering activations from background noise. These actively generated negative samples cannot be eliminated by the focal loss function. Therefore, we need to build a bridge between the multilevel extractor and the multibranch head network to renormalize the information flow and achieve the reorientation of the learning process. Naturally, we can implement this bridge with the help of RCs abstracted from KDN.

We then generalize the RCs to FPN-based multibranch detectors and verify that this approach solves the above problem well. The $n21C$ not only extends the multilevel and "divide-and-conquer" mechanism of the multilevel features of the FPN-based detectors to a wide range of scale-preferred tasks but also enables synergistic effects of multilevel features on the specific learning objectives. The $n21Cs$ assign appropriate subtasks to multiple detection branches for scale-preferred tasks. In addition, interfering activations, including those from the background and interfering negative samples, are greatly reduced and the detector redirects the learning direction to the correct one.

The RC renormalizes the information flow in all phases of network learning, rearranging the feature flow in the forward propagation phase and reorienting the gradient flow in the backpropagation phase. The information flow in both phases together determines the learning direction of the network. The design mechanism of RC lies in its "synergistic effect" on multiscale features and its "focus" on the dominant objective.

To meet different task requirements, the RCs can be seen as a customized plug-in module to renormalize the information flow of the network and redirect the learning process. The customized plug-in property means that there is flexibility in designing different connection strengths, embedding positions (between the feature extractor and the head in which detection branch or branches), and the number of RCs so that the "synergistic focusing" mechanism can be integrated into the detector and adapted to different task requirements.

Extensive experiments of 17 well-designed detection architectures embedded with $n21Cs$ on five different levels of scale-preferred tasks (including a newly released high-difficulty scale-preferred dataset and scale-diversified tasks) validate the renormalization effect of the RC approach $n21Cs$, especially the simplest linear form E421C. And it achieves real-time detection without additional parameters, but with higher accuracy. We hope that this lightweight and generalizable RC will bring some improvements to scale-preferred object detection research and applications.

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The learning process of RC differs from the loss reduction and optimization process of existing deep learning networks. In future work, we will further deepen the training process of $n21$Cs on small object detection tasks. In addition, we can continue to improve the representational capacity of the detector to further enhance its performance.

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